Spatio-temporal wall-actuation modes and cycle-mean viscous dissipation in a lid-driven cavity

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Marked-up manuscript. This file is the revised manuscript with the changes made relative to the previous version marked. Text in blue is new or rewritten; text in red belonged to the previous version and has been removed or replaced. Obsolete figures and tables of the previous version are not reproduced here; the corresponding changes are itemised in the response to the editor. Because the physics-first version is a reorganisation of the paper, most paragraphs carry a mark.

Abstract

At fixed actuation energy, the lid-driven cavity admits controlled states whose cycle-mean dissipation differs by up to a factor of three, and the state selected most often by an optimizer is not the least dissipative. This study asks which state a physics-informed control of the two-dimensional cavity selects and how reproducibly. A physics-informed neural network learns the lid velocity in a truncated spatio-temporal Fourier basis, minimizing viscous dissipation at prescribed actuation energy for Reynolds numbers between one hundred and one thousand. At matched energy the first-harmonic control dissipates less than a uniform lid, whereas the second-harmonic control dissipates more. The sign of this ordering holds on three grids and against two reference lids; only its magnitude is reference-dependent. At the same lid speed, by contrast, it lowers dissipation by seventy-six percent at Reynolds number five hundred and by eighty-two percent at one thousand, so the two comparisons answer different questions. Repeated independent trainings yield three converged clusters: a stationary second-harmonic state, a time-dependent state, and a residual mixture. A full-spectrum audit adds a first-harmonic class that is empty under variance regularization. Recovery frequencies depend on Reynolds number, energy, initialization, loss weights, and control representation. The variance-regularization weight selects among these states without altering the Navier–Stokes problem, so the second-harmonic state is regularization-selected and suboptimal at fixed energy: existence, accessibility, and reproducibility do not imply energetic optimality.

Keywords: physics-informed neural networks; active flow control; lid-driven cavity; modal branch selection; lattice-Boltzmann method; reproducibility.

Nomenclature

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Roman symbols	
AR	cavity aspect ratio, L_x/L_y
C_3	spectral rank concentration, sum of the three largest control-energy fractions
D	instantaneous viscous dissipation rate, $(2/Re)\int_\Omega \mathbf{S}:\mathbf{S}\,d\Omega$
E_{act}	instantaneous actuation energy of the lid profile, $\int_0^L U_{\mathrm{lid}}^2\,dx$
E_r	realized cycle-averaged actuation energy of a transferred control
E^*	prescribed target actuation energy
f_1, f_2, f_t	first-harmonic, second-harmonic, and explicitly time-dependent control-energy fractions
H_N	normalized branch-selection entropy
K	kinetic energy, $\frac{1}{2}\int_\Omega \lvert\mathbf{u}\rvert^2\,d\Omega$
L, L_x, L_y	cavity side, streamwise and wall-normal lengths
N	number of lattice nodes per cavity side (LBM resolution)
P_{lid}	mechanical power injected through the lid, $\int_0^L U_{\mathrm{lid}}\tau_w\,dx$
R_K	ratio of fluctuating to mean kinetic energy
Re	Reynolds number, $U_{\mathrm{ref}}L/\nu$
$\mathbf{S}$	strain-rate tensor, $(\nabla\mathbf{u}+\nabla\mathbf{u}^{\mathsf{T}})/2$
t, T	time and forcing period
$\mathbf{u}=(u,v)$	velocity vector and its streamwise/wall-normal components
u'	fluctuating velocity, $\mathbf{u}-\langle\mathbf{u}\rangle$
u'_{rms}	root-mean-square temporal fluctuation at a fixed point
U_{lid}	lid velocity profile (the actuation)
U_{ref}	reference velocity, the amplitude of the uniform lid
U_{SR}	symbolic-regression reconstruction of the mean lid profile
x, y	streamwise and wall-normal coordinates
Greek symbols and branch symbols	
B_1, B_2, B_3	empirical branch classes (second-harmonic dominated, time-dependent dominated, residual)
B_{first}	first-harmonic branch class of the full-spectrum audit
ε	integrated viscous dissipation accumulated over an LBM sampling window
λ_{var}	weight of the variance-regularization term of the training loss
ν	kinematic viscosity
τ_w	wall shear stress on the moving lid
ψ	stream function
ω	vorticity
ω_{SR}	spatial frequency returned by the symbolic fits
Ω	cavity domain
Abbreviations	
GCI	grid-convergence index
LBM	lattice-Boltzmann method
LOCO	leave-one-Reynolds-condition-out (cross-validation)
MRT	multiple-relaxation-time
PINN	physics-informed neural network
RMSE	root-mean-square error
SR	symbolic regression

INTRODUCTION

Active flow control seeks to modify fluid motion through externally imposed actuation in order to achieve a prescribed dynamical or energetic objective, including drag and lift modification, separation delay, mixing enhancement, and viscous dissipation reduction . The difficulty of this problem arises from the nonlinear and high-dimensional character of the Navier–Stokes equations: even when the actuator is parameterized by a small number of degrees of freedom, its interaction with the flow produces multiple competing responses, strongly non-convex optimization landscapes, and, in many configurations, multiple stable states. Reviews of flow control and machine-learning-based fluid mechanics have consequently emphasized the importance of reduced representations, interpretable flow structures, and systematic treatment of nonlinear effects rather than relying solely on scalar performance indicators . Methodological rigor in reporting convergence outcomes is especially important when physics-informed or data-driven methods are used, since a single converged solution may not represent the full landscape of accessible outcomes, and its physical

significance can only be assessed by characterizing how robustly it is reached and whether it persists under independent numerical scrutiny.

Low-dimensional representations provide a natural route to tractable flow-control problems. Proper orthogonal decomposition, balanced truncation, dynamic mode decomposition, resolvent analysis, and related techniques have become standard tools for identifying dynamically important flow structures and for constructing reduced-order models amenable to analysis and control . In control applications, such representations substantially compress the optimization problem while retaining structures relevant to the dominant flow response. Reduced-order optimal-control formulations have been applied to DNS-based flow-control calculations , and reinforcement-learning methods have extended this framework to complex geometries including bluff-body drag reduction . The lid-driven cavity occupies a canonical position in these investigations: its geometry and boundary conditions are simple, yet its recirculating flow structure and Reynolds-number dependence provide a non-trivial test of numerical and physical hypotheses, and high-quality benchmark solutions are available for independent verification . At moderate and high Reynolds numbers the uncontrolled cavity develops a rich sequence of flow structures—corner eddies, secondary recirculations, and an unsteady shear layer along the moving wall—whose faithful resolution places stringent demands on the numerical method and on the interpretation of any optimization outcome .

The energetic physics of this configuration is what makes it a useful test bed. The lid does work on the fluid at a rate set by the product of the lid velocity and the wall shear stress, and that work is balanced by viscous dissipation, which is concentrated in the lid shear layer, in the primary recirculation, and near the downstream corner where the boundary layer separates. Reshaping the lid profile therefore changes two things at once: the mechanical power injected into the cavity, and the spatial distribution of the strain rate at which that power is dissipated. A control that weakens the corner separation can reduce dissipation while injecting less power, so that a comparison at fixed lid velocity and a comparison at fixed actuation energy answer different physical questions. The first asks whether a given wall motion produces a more efficient flow; the second asks whether the same energetic budget is spent better. Only the second is a statement about control efficiency, and it is the one adopted here. The question addressed below is thus a physical one: among the flow states that a low-dimensional wall actuation can reach at a fixed energy budget, which one does the optimization select, and is that state the least dissipative of them.

Physics-informed neural networks (PINNs) offer an alternative route to reduced representations . Rather than learning a flow response exclusively from precomputed simulation data, PINNs incorporate the governing differential equations, boundary conditions, and physical constraints directly into the optimization objective through automatic differentiation of the network residuals. This formulation is particularly attractive for control identification because the flow field and the actuation parameters are optimized simultaneously, allowing the control itself to become an object of analysis rather than merely an output of a separate surrogate model. Machine learning has been applied broadly to fluid mechanics for model reduction, turbulence closure, and optimization , and physics-informed variants have demonstrated promising results for laminar, transitional, and turbulent flows. However, the physics-informed formulation does not remove the underlying non-convexity of the optimization problem. Different random initializations, network capacities, loss-weight configurations, regularization terms, or control parameterizations can lead to qualitatively different converged states. The repeated appearance of a particular low-dimensional structure cannot, by itself, establish that the structure is unique, globally optimal, or independent of the chosen representation.

This issue motivates a distinction that is rarely made explicit in physics-informed flow-control studies: the distinction between the *existence* of a controlled branch, its *accessibility* to a particular optimization procedure, and its *global optimality*. A control branch may be physically realizable and repeatedly recovered while remaining only one member of a set of competing converged states. Conversely, a branch that is infrequently recovered may not be physically less relevant; it may simply be recovered less often under the adopted optimization formulation. The control

parameterization introduces a further layer of conditioning: a finite basis does not merely reduce the number of degrees of freedom, but defines the subspace of control space in which the optimizer searches. An apparently low-dimensional control structure can therefore reflect both the dynamics of the controlled flow and the geometry of the chosen optimization space . Subcritical non-convex landscapes in fluid mechanics have been characterized for variational and adjoint-based methods , and resolvent analysis has shown how structural assumptions can influence perceived optimal structures . A quantitative framework for distinguishing these contributions is therefore essential before drawing physical conclusions from a repeatedly observed modal structure.

The present study addresses this branch-selection problem for distributed active boundary control of the two-dimensional lid-driven cavity. We formulate a physics-informed optimization in which the Navier–Stokes equations constrain the flow while a prescribed actuation-energy level and a dissipation-related objective determine the control. The actuation is represented in a truncated spatio-temporal Fourier basis, and a parametric neural representation identifies controls over the investigated Reynolds-number interval. Rather than treating the dominant modal structure as an assumed optimum, we explicitly test its recoverability, multiplicity, and parameter dependence. Independent training realizations estimate empirical branch-selection frequencies

$$\widehat{P}(B_i \mid Re, E^*, \lambda_{\mathrm{var}}) \,, \tag{1}$$

while systematic variations of Re, E^*, network capacity, control-space truncation, cavity aspect ratio, loss weighting, and control basis determine when the observed branches persist or change. The resulting branch structure is then subjected to an independent numerical test: selected controls are transferred without retraining to an independent multiple-relaxation-time lattice-Boltzmann (MRT-LBM) solver, providing a completely independent discretization and implementation. Grid refinement and benchmark comparisons quantify the associated numerical uncertainty, and symbolic regression provides a compact analytical approximation enabling assessment of cross-condition generalization. The aim is a physical account of how a low-dimensional wall motion selects among competing cavity states, and of what that selection costs in dissipation.

The central questions of this study are therefore: (i) Does a distributed physics-informed control problem repeatedly select a low-dimensional modal branch over a range of flow and actuation parameters? (ii) Do independent initializations reveal competing branches with comparable optimization performance? (iii) How does the variational structure of the loss redistribute the empirical accessibility of those branches? (iv) Does the observed modal structure survive changes in the control representation? (v) Can the resulting branches be independently realized and distinguished by a conventional numerical flow solver? Answering these questions provides a more stringent interpretation of low-dimensional structures obtained from physics-informed optimization and places modal concentration within the broader framework of nonlinear branch selection in controlled fluid flows. The paper follows the physics from the flow state to its energy budget. Section **??** formulates the controlled cavity, the actuation energy that fixes the comparison, and the optimization. Section **??** maps which flow states are reached across the (Re, E^*) plane. Section **??** isolates what controls their accessibility, and Sec. **??** tests whether they survive changes of capacity, dimension, geometry, and basis. Section **??** transfers the selected states to an independent solver and measures their dissipation, which Sec. **??** interprets in energetic terms. Section **??** condenses the dominant mean profile into an algebraic form, and Sec. 8 states the conclusions.

PROBLEM FORMULATION AND PHYSICS-INFORMED OPTIMIZATION

The objective is to examine how a distributed boundary-control problem can admit competing low-dimensional actuation branches and how their accessibility changes with the physical and variational parameters of the identification procedure. Before introducing the physics-informed optimization, the controlled flow problem and the admissible actuation space must be specified independently of the neural network representation. All notation is consistent throughout the paper; additional implementation details are provided in the Supplementary Material.

Abstract

The spatial and temporal organization of wall actuation determines how mechanical input is distributed across the flow structures of a driven cavity, yet the resulting viscous-dissipation cost cannot be inferred from the actuation spectrum or from the time-mean velocity field alone. This study examines the relationship between wall-actuation structure, hydrodynamic response, and cycle-mean viscous dissipation in a two-dimensional lid-driven cavity. Stationary harmonic profiles and three physics-informed actuation branches are transferred to an independent multiple-relaxation-time lattice-Boltzmann solver and compared at a common quadratic actuation norm. For stationary controls at $Re = 500$, the first spatial harmonic reduces dissipation by 17.7% relative to a corner-regularized uniform-lid reference, whereas the second harmonic and the mean profile of the second-mode-dominated branch increase it by 36.8% and 44.3%, respectively. Grid-refinement and local strain-rate analyses associate this ordering with the redistribution of dissipation between corner shear layers and the cavity interior. For time-dependent branches, the cycle-mean dissipation differs markedly from the dissipation of the time-mean flow. In particular, the temporally dominated branch has a cycle-mean dissipation approximately 3.1 times that of the predominantly stationary branch, despite its much weaker mean-flow contribution. The results demonstrate that the energetic cost of wall actuation depends on both the spatial organization and the temporal fluctuations of the resulting flow, and that physical performance must be assessed using cycle-averaged dissipation rather than modal accessibility or mean-field diagnostics alone.

Keywords: lid-driven cavity; wall actuation; viscous dissipation; energy budget; mean-flow diagnostics; mode selection

1 Introduction

Wall actuation is one of the few ways in which the vortical structure of a confined recirculating flow can be modified without changing its geometry. In a lid-driven cavity the entire vortex topology is generated by the tangential motion of a single boundary, so any spatial structure imposed on that boundary acts directly on the mechanism that sustains the flow. The cavity is therefore a natural configuration in which to ask a question that is central to flow control but rarely posed in energetic terms: when two different actuations require the same control effort, which one produces the less dissipative flow, and why? The canonical cavity solution and its benchmarks are well established [1, 2, 3, 4, 5], and the flow has long served as a test bed for control strategies [6, 7], yet its dissipation budget under distributed wall actuation has received comparatively little attention.

The energetic mechanism is twofold, and the two parts can act in opposite directions. Changing the wall velocity changes the mechanical power injected into the fluid, $P_{\mathrm{lid}} = \int_0^1 U_{\mathrm{lid}} \tau_w \, dx$, where τ_w is the wall shear stress; it also changes the spatial distribution of the strain-rate tensor $\mathbf{S}$, and with it the partition of viscous dissipation between the lid shear layer, the side-wall layers, the cavity interior and the two upper corners, where a discontinuous lid produces a stress singularity. A profile can therefore reduce the peak strain at the corners while increasing the strain everywhere else, and the net effect is not predictable from the shape of the actuation alone. The distinction matters because the two natural comparison protocols answer different questions. Comparing profiles at the same lid-velocity scale [32] measures what a given actuator amplitude achieves; comparing them at the same quadratic actuation norm $E_{\mathrm{act}} = \int_0^1 U_{\mathrm{lid}}^2 \, dx$ measures what a given control effort achieves. The two rankings need not agree, and only the second is a statement about efficiency. A further distinction, specific to time-dependent actuation, is that the dissipation of the time-mean field is not the time mean of the instantaneous dissipation: because D is quadratic in the velocity gradients, cycle-averaged dissipation always exceeds the dissipation of the mean flow, and the gap can be orders of magnitude when the fluctuations dominate [23, 24].

Spatially structured and time-periodic actuations can also support more than one hydrodynamic response. A periodic boundary-value problem of this type admits several coexisting branches, and which one is reached depends on the formulation, the regularization and the initialization of the identification procedure. A preceding study by Beniaiche [32] investigated the identification and independent numerical verification of conditional modal collapse in physics-informed active control

of the lid-driven cavity. It established that the identified actuation is not characterized by a unique, universally selected modal structure: predominantly stationary second-harmonic controls and strongly time-dependent alternatives can be recovered depending on initialization, regularization, and control parameterization. Independent MRT-LBM simulations further demonstrated that these identified controls can produce physically distinct cavity responses. The emphasis of that work was on branch identification, conditional accessibility, and independent numerical reproducibility. The present study addresses the complementary fluid-mechanical question of how these spatial and temporal actuation structures affect the viscous-dissipation budget of the resulting flow. Rather than treating modal recovery as a performance criterion, we quantify the dissipation of stationary and periodic responses under a common quadratic actuation norm, distinguish the dissipation of the time-mean field from the cycle-mean dissipation, and identify the strain-rate distributions responsible for the observed energetic differences.

The question addressed in this paper is accordingly:

> How does the spatio-temporal structure of wall actuation determine the cycle-mean viscous dissipation of the resulting cavity flow at a prescribed actuation-energy level?

The sequence of the paper follows that question. Section 2 fixes the physical and energetic framework, and in particular separates the quadratic actuation norm from the mechanical power transmitted through the wall. Section 3 defines the stationary modes and the three identified branches that are compared. Section 4 describes how the actuations are identified and how the flows are computed independently. Section 5 reports the energetic response of the stationary modes, Section 3.2 the cycle-mean response of the time-dependent branches, and Section 7 interprets both in terms of local strain and fluctuating gradients. Section 8 concludes.

2 Physical formulation and energetic framework

2.1 Controlled cavity and governing equations

We consider a two-dimensional incompressible lid-driven cavity in which the upper boundary is subjected to a spatially and temporally distributed tangential actuation. The lower and lateral walls remain stationary with no-slip conditions. The lid velocity is not prescribed *a priori* as a uniform profile but is treated as a control variable drawn from a finite-dimensional admissible space. This is essential for the subsequent branch analysis: the branches investigated here are not solutions of different physical problems, but different control-flow realizations admitted by the same governing equations, boundary geometry, and actuation constraints.

After nondimensionalization by the cavity height L_y^{dim}, the wall-normal extent is unity and the streamwise extent equals the aspect ratio $AR = L_x^{\mathrm{dim}}/L_y^{\mathrm{dim}}$:

$$0 \le x \le L_x, \quad 0 \le y \le 1, \quad L_x = AR. \tag{2}$$

The reference square cavity corresponds to $L_x = 1$. Velocities are nondimensionalized by a reference lid velocity U_0, time by L_y^{dim}/U_0, and pressure by ρU_0^2. The Reynolds number is

$$Re = \frac{U_0\, L_y^{\mathrm{dim}}}{\nu}, \tag{3}$$

where ν is the kinematic viscosity. Note that U_0 is a scaling velocity, not a requirement that the lid remain uniform; the distributed control U_{lid} may change sign locally provided it remains within the admissible control space and satisfies the prescribed energy constraint.

Two remarks delimit what this boundary condition claims. A tangential velocity prescribed pointwise along the upper wall is an idealization of actuation hardware that exists rather than a single rigid lid: an array of independently driven wall segments reproduces any profile $U_{\mathrm{lid}}(x)$ up to the segment pitch, and wall-bounded momentum injection without moving parts is routinely

produced by dielectric-barrier-discharge plasma actuators, whose near-wall body force is equivalent, in the time-averaged sense used here, to a distributed tangential wall velocity . What the formulation does *not* assert is that one rigid translating wall could impose a sign-changing profile; the energy constraint E^* of Sec. **??** is what keeps the commanded effort of the distributed actuator in a physically meaningful range. Second, the optimization is posed in the time-dependent formulation even though the reference outcome is quasi-stationary. A steady formulation restricts the admissible set to time-independent lid profiles and would exclude by construction the branch whose dominant content is temporal (B_2), so the branch census reported here is only defined in the unsteady setting; the quasi-stationary B_1 outcome is not an artifact of that choice, because it is recovered as a stationary state by the independent LBM verification of Sec. **??**.

The controlled velocity field $\mathbf{u} = (u, v)$ and kinematic pressure p satisfy the dimensionless incompressible Navier–Stokes equations

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla)\mathbf{u} = -\nabla p + \frac{1}{Re}\nabla^2 \mathbf{u}, \quad \nabla \cdot \mathbf{u} = 0. \tag{4}$$

The stationary boundaries satisfy the no-slip condition

$$\mathbf{u} = \mathbf{0}, \quad x = 0,\ x = L_x,\ y = 0, \tag{5}$$

and the controlled upper boundary satisfies

$$u(x, 1, t; Re) = U_{\mathrm{lid}}(x, t; Re), \quad v(x, 1, t; Re) = 0. \tag{6}$$

The problem is nonlinear for two reasons: the Navier–Stokes convective term, and the boundary condition ?, which is itself selected from a parameterized control space. The optimization therefore couples the selection of a boundary actuation to the nonlinear flow response that the same actuation generates.

Finite-dimensional spatio-temporal control

The lid actuation is represented in a truncated spatio-temporal Fourier basis. The sine basis in x enforces $U_{\mathrm{lid}}(0, t; Re) = U_{\mathrm{lid}}(L_x, t; Re) = 0$, preventing discontinuous tangential velocity at the upper corners for every tested aspect ratio. The cosine basis in t allows the $n = 0$ terms to represent purely stationary actuation, the most directly interpretable patterns physically. The lid velocity is expanded as

$$U_{\mathrm{lid}}(x, t, Re) = \sum_{m=1}^{N_x} \sum_{n=0}^{N_t - 1} a_{mn}(Re) \sin\left(\frac{m\pi x}{L_x}\right) \cos(n\pi t). \tag{7}$$

The temporal coordinate in Eq. ? is the normalized PINN time: the residual is imposed on $t \in [0, 1]$, and every cosine factor is evaluated on that interval. Because the fundamental factor is $\cos(\pi t)$, the control is periodic with period $T = 2$ in these units, the n-th temporal harmonic having period $2/n$. The control is therefore exported on the full fundamental period $t \in [0, 2]$, and the LBM resolves that period with 256 steps, i.e. 128 steps per unit of PINN time. All transient LBM runs of the branch comparison start from rest ($\rho = 1$, $\mathbf{u} = \mathbf{0}$), with the lid switched on at $t = 0$; at least 200 forced periods are discarded as transient, a drift test on the cycle-averaged dissipation then fixes the end of the settling phase, and 64 consecutive periods are retained for the statistics. The same initialization, transient rule and sampling window are used for B_1, B_2 and B_3. All modal labels use the physical (one-based) convention: the dominant stationary second spatial harmonic corresponds to $(m, n) = (2, 0)$. The baseline control space uses $(N_x, N_t) = (6, 5)$, giving 30 nominal coefficients. Control-space sensitivity examines

$$(N_x, N_t) \in \{(4, 3), (6, 1), (6, 3), (6, 5), (8, 5), (8, 9)\}, \tag{8}$$

spanning 6 to 72 nominal modes. The modal energy of coefficient (m, n) is

$$\mathcal{E}_{m0} = \tfrac{1}{2} a_{m0}^2, \quad \mathcal{E}_{mn} = \tfrac{1}{4} a_{mn}^2 \quad (n \geq 1), \tag{9}$$

where the factors account for temporal basis normalization ($\langle\cos^2(0)\rangle = 1$, $\langle\cos^2(n\pi t)\rangle = \frac{1}{2}$ for $n \geq 1$). The normalized modal fraction is

$$f_{mn} = \frac{\mathcal{E}_{mn}}{\sum_{i=1}^{N_x}\sum_{j=0}^{N_t-1}\mathcal{E}_{ij}}, \qquad \sum_{m=1}^{N_x}\sum_{n=0}^{N_t-1} f_{mn} = 1. \tag{10}$$

The stationary second-harmonic fraction is $f_2 \equiv f_{20}$, and the total time-dependent fraction is $f_t = \sum_{m=1}^{N_x}\sum_{n=1}^{N_t-1} f_{mn}$. The realized actuation energy is

$$E_r = \langle U_{\mathrm{lid}}^2\rangle_{s,t} = \frac{1}{T}\int_0^T\int_0^1 U_{\mathrm{lid}}^2(s,t;Re)\,ds\,dt, \quad s = \frac{x}{L_x}, \tag{11}$$

and the optimization is conditioned on $E_r \approx E^*$. Modal-energy quadrature conventions are detailed in Supplementary Sec. S2.

Parametric PINN formulation

The flow field and Reynolds-dependent control coefficients are represented simultaneously by a parametric neural network with inputs $(x, y, t, \widetilde{Re})$, where

$$\widetilde{Re} = \frac{Re - 550}{450} \in [-1, 1], \tag{12}$$

maps the training interval $[100, 1000]$ to $[-1, 1]$. A single trained network represents the flow-control pair over the entire Reynolds-number range, enabling efficient evaluation of the branch map without repeated retraining at each condition. The flow variables (u, v, p) and control coefficients $\{a_{mn}(\widetilde{Re})\}$ are the network outputs. Two subnetworks are employed: a flow network with 6 hidden layers of 64 neurons each using tanh activation (21 379 trainable parameters), and a control network with 4 hidden layers of 32 neurons each (3 614 parameters). All computations are performed in double-precision arithmetic, and all physical residuals are evaluated by automatic differentiation. Full architecture specifications are provided in Supplementary Sec. S1.

Optimization objective and training procedure

The multi-objective loss is

$$\mathcal{L} = \lambda_{\mathrm{PDE}}\mathcal{L}_{\mathrm{PDE}} + \lambda_{\mathrm{BC}}\mathcal{L}_{\mathrm{BC}} + \lambda_{\mathrm{diss}}\mathcal{L}_{\mathrm{diss}} + \lambda_{\mathrm{ctrl}}\mathcal{L}_{\mathrm{ctrl}} + \lambda_{\mathrm{var}}\mathcal{L}_{\mathrm{var}} + \lambda_{Re}\mathcal{L}_{Re}, \tag{13}$$

with reference weights $\lambda_{\mathrm{PDE}} = 1$, $\lambda_{\mathrm{BC}} = 15$, $\lambda_{\mathrm{diss}} = 0.05$, $\lambda_{\mathrm{ctrl}} = 200$, $\lambda_{\mathrm{var}} = 1$, $\lambda_{Re} = 2$. The individual loss components are:

PDE residual loss: $\mathcal{L}_{\mathrm{PDE}} = \langle R_u^2 + R_v^2 + R_c^2\rangle_{\chi_{\mathrm{int}}}$, where R_u, R_v, R_c are momentum and continuity residuals evaluated at $N_{\mathrm{phys}} = 4\,000$ interior collocation points resampled uniformly at each epoch.

Boundary condition loss: $\mathcal{L}_{\mathrm{BC}} = \langle u^2 + v^2\rangle_{\Gamma_{\mathrm{stat}}} + \langle[u - U_{\mathrm{lid}}]^2 + v^2\rangle_{\Gamma_{\mathrm{top}}}$, where $\Gamma_{\mathrm{stat}} = \{x = 0\} \cup \{x = L_x\} \cup \{y = 0\}$. The weight $\lambda_{\mathrm{BC}} = 15$ enforces boundary conditions strongly relative to the interior residual.

Viscous dissipation objective:

$$\mathcal{L}_{\mathrm{diss}} = \int_\Omega \frac{1}{Re}\left[u_x^2 + v_y^2 + \tfrac{1}{2}(u_y + v_x)^2\right] d\Omega = \int_\Omega \nu\, S{:}S\, d\Omega. \tag{14}$$

This term drives the optimizer toward low-dissipation actuation patterns and constitutes the physical objective of the control problem. Here $S_{ij} = \frac{1}{2}(\partial_i u_j + \partial_j u_i)$ is the strain-rate tensor, so $\mathcal{L}_{\mathrm{diss}}$ equals exactly half the standard integrated dissipation rate $\varepsilon = 2\nu S{:}S$ reported by the independent LBM diagnostic; the factor of two is absorbed by λ_{diss} and does not affect the optimum.

Actuation energy penalty: $\mathcal{L}_{\mathrm{ctrl}} = (E_r - E^*)^2$. The large weight $\lambda_{\mathrm{ctrl}} = 200$ enforces the energy budget tightly, ensuring that competing converged solutions are compared at the same

actuation energy level and that the modal competition is not trivially explained by different energy budgets.

Variance regularization:

$$\mathcal{L}_{\text{var}} = -\log\Big[\text{Var}(U_{\text{lid}}) + 10^{-5}\Big]. \tag{15}$$

This term penalizes spatially uniform or near-uniform lid profiles by rewarding higher spatial variance. It plays a critical role in breaking degeneracy toward trivially flat controls and is the primary determinant of branch selection, as demonstrated in Section **??**.

Reynolds-separation regularization:

$$\mathcal{L}_{Re} = \max\left[0,\ \delta - \frac{1}{N_{\text{probe}}}\sum_{i=1}^{N_{\text{probe}}}\left(U_{\text{lid}}(x_i, t_0, Re_{\min}) - U_{\text{lid}}(x_i, t_0, Re_{\max})\right)^2\right], \tag{16}$$

with $N_{\text{probe}} = 20$, $t_0 = 0.5$, $\delta = 0.01$, preventing the parametric network from collapsing to a Reynolds-independent control.

No initial-condition term appears in Eq. ?, and this omission is deliberate. Because the flow and the control are expanded in a temporal Fourier basis and the objective is a cycle-averaged dissipation at a prescribed actuation energy, the network is formulated as a periodic space–time boundary-value problem: it seeks a statistically stationary, time-periodic state compatible with the wall boundary conditions, not the transient evolution from a prescribed initial field. The word "initialization" in this work therefore refers to the initialization of network parameters, never to a flow initial condition. The PINN results consequently characterize which periodic state a training protocol reaches, not how long a physical flow takes to reach it from rest, and they carry no phase information relative to a physical initial time. The independent LBM calculations used for verification do start from rest and discard at least 200 periods before sampling (Sec. **??**), so initial-value behavior is checked there rather than assumed for the PINN.

Training consists of a 300-epoch pre-training phase in which only the flow network is updated with control coefficients frozen at zero, establishing an initial physically consistent flow representation. A 3 000-epoch joint optimization phase follows using the Adam optimizer with initial learning rate $\eta_0 = 8 \times 10^{-4}$ and exponential decay by factor 0.995 per 100 epochs. Collocation points are resampled at every epoch. Complete sampling details are provided in Supplementary Sec. S1.

Branch classification and reproducibility protocol

Converged neural realizations are classified into empirical branch classes according to their modal signature computed from Eqs. ?–?. The rules are applied in the order below, and f_1 denotes the stationary first-harmonic fraction, i.e. the energy carried by $(m, n) = (1, 0)$:

- **Branch B_{first} — Stationary first-mode dominated:** $f_1 > 0.5$. This class was added after the modal audit of the full spectrum (Supplementary Sec. S2.3). It is *empty* in the 185-realization campaign, where f_1 never exceeds 12.2% (and 9.6% within B_3); it is populated only by the $\lambda_{\text{var}} = 0$ ablation, where f_1 reaches 87–99%.
- **Branch B_1 — Stationary second-mode dominated:** $f_2 > 2f_t$ and $f_2 > 0.5$. Physical harmonic $(m, n) = (2, 0)$. Ensemble mean $\overline{f_2} = 85.3\%$, $\overline{f_t} = 7.4\%$.
- **Branch B_2 — Temporally dominated:** $f_t > 2f_2$ and $f_t > 0.5$. Physical harmonic $(m, n) = (1, 1)$. Ensemble mean $\overline{f_2} = 7.5\%$, $\overline{f_t} = 88.1\%$.
- **Branch B_3 — Residual class:** neither dominance condition holds. Ensemble mean $\overline{f_2} = 34.8\%$, $\overline{f_t} = 43.9\%$, with only $\overline{f_1} = 3.0\%$ on the first spatial harmonic.

The classification is therefore based on modal *dominance* (a factor of two between the two modal families) rather than on an absolute cut-off, so that a realization is assigned to the class that

carries the control energy. The dominance factor 2 and the floor 0.5 are fixed a priori and not tuned to the data; the sensitivity of the pooled frequencies to both is quantified in Supplementary Sec. S2.3 and is below 3 percentage points for the two dominant classes. Adding B_{first} leaves the 185-realization labels unchanged, because no campaign realization comes close to the $f_1 > 0.5$ condition (Table **??**); the extension therefore corrects a naming defect without altering the reported frequencies, confidence intervals or figures. What B_3 actually contains is reported separately: it is not a homogeneous class. Of its 26 members, 15 are dominated by the temporal family ($f_t = 0.37$–0.64), 8 by the stationary second harmonic ($f_2 = 0.39$–0.63) and 3 by stationary harmonics of order $m \geq 3$ (0.74–0.84); none is dominated by the first harmonic (Table **??**). The empirical selection frequency over N_s independent seeds is $\widehat{P}(B_k) = N_k/N_s$. Selection diversity is measured by normalized Shannon entropy $H_N = -\frac{1}{\ln 3}\sum_k \widehat{P}(B_k)\ln\widehat{P}(B_k)$, which equals zero for deterministic single-branch selection and unity for a uniform distribution over the three classes populated in the campaign. Detailed implementation parameters and the cell-aggregated campaign data are provided in the Supplementary Material.

GLOBAL BRANCH-SELECTION LANDSCAPE

Broad (Re, E^*) campaign

The broad campaign evaluated 185 independent training runs across 30 parameter-space cells ($Re \in \{100, 300, 500, 700, 900, 1000\}$, $E^* \in \{0.1, 0.25, 0.5, 1, 2\}$), with at least 5 seeds per cell and up to 20 at selected cells of particular interest. Figure **??** presents the resulting empirical branch-frequency map $\widehat{P}(B_k)$, and Table **??** summarizes the global realization statistics. The cell-aggregated campaign dataset—realization counts, modal statistics and empirical class frequencies for each of the 30 cells—is provided in Supplementary Sec. S3 and Supplementary Table S3. The realized energy of these runs is not exactly the prescribed one: the cell mean of $\langle E_r\rangle/E^*$ stays within 3% of unity for $E^* \geq 0.25$ but every realization of the $E^* = 0.1$ column overshoots, with a mean of 1.209, so the lowest-energy column is explored at a slightly larger budget than prescribed (Supplementary Fig. S1).

Across the 185 runs, B_1, B_2, and B_3 account for 42.2%, 43.8%, and 14.1% of solutions, respectively. The near balance between B_1 and B_2 is the most significant feature of the global landscape: while the stationary second-harmonic branch is frequently accessible, it does not dominate the ensemble to the point of being the unique or near-deterministic outcome. The residual branch B_3 is the least frequently selected, arising primarily at elevated actuation energies $E^* \geq 1$ and higher Reynolds numbers. The branch frequencies exhibit a clear Re-dependence: B_1 is more readily accessible at moderate Re around 500, while at $Re \in [100, 300]$, B_2 often dominates. This suggests that the relative accessibility of the two competing outcome clusters varies systematically with the flow regime.

Continuous modal structure in observable space

Figure **??** illustrates the distribution of all 185 realizations in the two-dimensional (f_2, f_t) observable space. Three clusters of training outcomes are clearly visible, separated by low-density gaps. The B_1 cluster concentrates near the $f_t \approx 0$ axis at high f_2; the B_2 cluster concentrates near the $f_2 \approx 0$ axis at high f_t; and the B_3 cluster occupies an intermediate region. The separation is consistent with three empirically distinguishable classes of converged outcomes; the finite sample per cell limits the precision of frequency estimates and does not by itself establish the existence of dynamical attractors.

The separation of Figure **??** is stated with two scalars, but the modal composition behind it is equally distinct. Figure **??** shows the mean energy fraction $\overline{f}_{mn}$ of the six most populated harmonics of Eq. ?, averaged over the realizations of each class. B_1 concentrates 85.3% of the control energy in the stationary second harmonic $(m, n) = (2, 0)$ and 3.7% in the leading temporal harmonic $(1, 1)$; B_2 inverts these numbers, with 75.4% on $(1, 1)$ against 7.5% on $(2, 0)$; B_3 has no single dominant harmonic, its largest stationary component being 34.8% on $(2, 0)$ against 43.9% in the temporal family as a whole, and it still carries 15.7% on the third stationary harmonic $(3, 0)$ and 8.7% on $(2, 1)$, six and nine times the corresponding B_1 values. The temporal content of B_2 is itself broader than its leading mode: $\overline{f_t} = 88.1\%$ against 75.4% on $(1, 1)$, so about 13% of the

budget is carried by higher temporal harmonics. The residual character of B_3 is therefore spectral as well as statistical, and the two dominance inequalities of Sec. **??** summarize a reorganization of the whole control spectrum rather than an arbitrary projection onto two numbers.

Conditional branch accessibility and non-determinism

A key observation from Figure **??** is that (Re, E^*) does not uniquely determine a single control class. At $Re = 500$, $E^* = 0.25$, a 15-seed sub-campaign yields $\widehat{P}(B_1) = 0.467$, $\widehat{P}(B_2) = 0.400$, $\widehat{P}(B_3) = 0.133$, demonstrating genuine multi-branch accessibility at a single operating point. At $Re = 1000$, $E^* = 1$, a 20-seed sub-campaign yields $\widehat{P}(B_1) = 0.55$, $\widehat{P}(B_2) = 0.20$, $\widehat{P}(B_3) = 0.25$. The mapping $(Re, E^*) \to \mathcal{B}$ is therefore set-valued: each operating point maps to an empirical distribution $\{\widehat{P}(B_1), \widehat{P}(B_2), \widehat{P}(B_3)\}$. Table **??** reports these frequencies with 95% Wilson score intervals; at these sample sizes the intervals are wide, so the ordering of frequencies within a cell should be read as a trend rather than a precise probability. The following sections investigate the conditions under which this distribution changes.

CONDITIONAL BRANCH ACCESSIBILITY

Variational dependence on λ_{var}

The effect of variance regularization was tested in a dedicated 90-run campaign ($N_s = 5$ seeds at 3 operating points, 6 values of $\lambda_{\text{var}} \in \{0, 0.25, 0.5, 1, 2, 5\}$). Table **??** and Figure **??** demonstrate that changing λ_{var} systematically redistributes empirical branch-selection frequencies while leaving the Navier–Stokes problem entirely unchanged. Complete seed-level results and finite-sample Wilson-score confidence intervals are given in Supplementary Sec. S4.

Figure **??** states the same redistribution on a single scale. H_N jumps from zero at $\lambda_{\text{var}} = 0$ to between 0.613 and 0.960 whenever a nonzero weight produces a mixture. The one exception is $(Re, E^*) = (700, 2)$ at $\lambda_{\text{var}} = 0.25$, where all five seeds remain B_{first}: the collapse of the unregularized optimizer is therefore not removed by an arbitrarily small penalty. The dependence is non-monotonic at all three operating points, and at $(Re, E^*) = (700, 2)$ the entropy falls back to 0.865 once $\lambda_{\text{var}} \geq 2$ because B_1 regains the majority of the seeds.

At $\lambda_{\text{var}} = 0$, all three operating conditions yield exclusively first-harmonic realizations of class B_{first} ($f_1 = 0.87$–0.99, $H_N = 0$): in the absence of spatial-variance regularization, the optimizer collapses to the lowest spatial mode without developing spatially structured patterns. Introducing any nonzero λ_{var} immediately activates structured patterns, shifting accessibility toward B_1 or B_2 depending on the operating point. The transition is non-monotonic: at $(Re, E^*) = (500, 0.25)$, B_1 frequency is maximized at intermediate $\lambda_{\text{var}} \approx 0.25$, whereas larger values shift the distribution toward B_2, reaching $\widehat{P}(B_2) = 0.6$ at $\lambda_{\text{var}} = 5$. At $(Re, E^*) = (700, 2)$, the opposite trajectory is observed: values in the range $[2, 5]$ favor B_1 even though small values favor B_{first} or B_2. These results demonstrate that variance regularization acts as an optimization landscape bias: it modifies which outcome cluster the optimizer converges to during training without modifying the underlying fluid-mechanics problem. This constitutes direct evidence that the branch structure observed in physics-informed control studies is at least partially a property of the variational formulation rather than solely of the physical problem.

Dependence on actuation energy

Target-energy sweeps at fixed $Re = 500$ over $E^* \in \{0.01, 0.05, 0.1, 0.25, 0.5, 1, 2\}$ reveal that the second-harmonic purity f_2 decreases monotonically as the prescribed actuation amplitude grows, while f_t correspondingly increases (Figure **??**). At the lowest energies ($E^* \leq 0.05$), the control budget is insufficient to explore temporal degrees of freedom, and the optimizer concentrates almost entirely on the second spatial harmonic ($f_2 \approx 93$–94%). At the reference condition $E^* = 0.25$, $f_2 = 92.2\%$ with $f_t = 1.3\%$. At $E^* = 2$, however, f_2 drops to 74.0% while f_t rises to 21.8%, indicating that at large actuation amplitudes the optimizer increasingly distributes control energy away from the second harmonic toward additional spatial and temporal components. This transition reflects the fact that at high energy the energy budget can be satisfied by many different modal mixtures, making the second-harmonic outcome less uniquely favoured. Extended energy sweep datasets are provided in Supplementary Sec. S5.

Initialization sensitivity

Ten independent random seeds at $Re = 500$, $E^* = 0.25$ produced 4 second-mode dominated (B_1), 4 temporally dominated (B_2), and 2 intermediate (B_3) realizations. Figure **??** illustrates the seed-to-seed outcomes in the (f_2, f_t) space. Despite qualitatively distinct modal structures, total training losses are tightly clustered within $[1.371, 1.394]$, a relative spread of only 1.7%. The B_1 and B_2 realizations achieve essentially equivalent loss values within the observed spread (relative spread 1.7%), consistent with competing clusters of converged outcomes with comparable loss under the adopted formulation. This multiplicity is the direct signature of a non-convex optimization problem with competing local minima, and it demonstrates that the selection of a particular branch by any single training run cannot serve as evidence of global optimality. The complete seed-level dataset is provided in Supplementary Sec. S5.

Loss-term ablation

Ablation experiments at $Re = 500$, $E^* = 0.25$ isolate the role of individual loss components. Table **??** reports results for single and dual term removals. Removing $\mathcal{L}_{\text{var}}$ alone causes f_2 to collapse from 92.2% to 0.2%—a complete modal-class shift from B_1 to the first-harmonic class B_{first}. The archived five-seed repetition of the $\lambda_{\text{var}} = 0$ configuration shows that the collapsed control is not a mixture but a nearly pure first harmonic: $f_1 = 87$–99%, $f_2 \leq 2.5\%$ and $f_t \leq 12\%$ across the three operating points tested (Supplementary Sec. S5.3, five seeds per point). The residual energy is therefore concentrated in the lowest spatial mode, not spread over the spectrum, which is what the earlier label B_3 obscured. Removing $\mathcal{L}_{Re}$ or $\mathcal{L}_{\text{diss}}$ individually leaves f_2 essentially unchanged at 92.2–92.3%. Dual ablations confirm that the shift induced by removing $\mathcal{L}_{\text{var}}$ is not amplified or modulated by simultaneously removing other terms. Reducing λ_{var} by half decreases second-mode concentration modestly to 89.3%, while doubling it increases it to 92.8%. These results establish that $\mathcal{L}_{\text{var}}$ is the critical loss term for B_1 selection, and that its action is not redundant with the dissipation or Re-separation terms. Additional ablation datasets are given in Supplementary Sec. S5.

ROBUSTNESS AND REPRESENTATION DEPENDENCE

Neural-network capacity

To assess whether the second-harmonic concentration is an artifact of the specific network architecture, three capacity variants were evaluated at $Re = 500$, $E^* = 0.25$: a small network (4×32), the reference network (6×64), and a large network (8×128). The second-harmonic fraction remains in the range $f_2 \in [89.5\%, 92.2\%]$ across all three configurations, a spread of only 2.7% (Fig. **??**(a)). The fact that a network roughly 40 times smaller than the large variant produces essentially the same modal concentration argues against the hypothesis that the second-harmonic structure is imposed by the inductive bias of the neural network. Detailed architecture-level results are reported in Supplementary Sec. S6.

Control-space dimension

Truncation sweeps across six control dimensions (Table **??**) show persistent second-mode dominance with $f_2 \in [87.3\%, 92.3\%]$. Figure **??**(b) confirms that expanding the basis to 72 modes does not eliminate the B_1 branch or substantially redistribute modal energy toward higher harmonics; additional modes remain largely unactivated. The second-harmonic amplitude A_2 varies only from 0.668 to 0.684 across all truncations. These results concern the robustness of the selection, not its physical efficiency: they show that the second-harmonic outcome is not an artifact of insufficient degrees of freedom, and that the modal concentration is stable against the control dimension in the range tested. They do not indicate a low-dissipation response. At the matched energy $E^* = 0.25$ of Table **??**, the stationary second harmonic is the more dissipative member of the class it is compared with, so expanding the basis strengthens the claim that the optimizer keeps selecting this structure, while the energetic assessment of Sec. **??** keeps it suboptimal. Truncation datasets are provided in Supplementary Sec. S6.

Cavity aspect ratio

Changing the cavity geometry ($AR \in \{0.5, 1, 2\}$) at $Re = 500$, $E^* = 0.25$ shows that the stationary second-harmonic signature remains dominant at the reference aspect ratio ($f_2 = 92.2\%$) and at the elongated geometry ($AR = 2$, $f_2 = 91.2\%$), but its concentration decreases notably at

$AR = 0.5$, where f_2 drops to 69.9% while f_t rises to 26.7% (Table **??**). The compressed geometry changes the relationship between the harmonic structure of the Fourier basis and the natural recirculation scales of the cavity: at $AR = 0.5$, the second harmonic has a half-wavelength of 0.25, shorter than the natural recirculation scale, and may be less efficient at redistributing the primary vortex structure. Detailed basis-construction data are provided in Supplementary Sec. S7.

Control-basis dependence

Replacing the Fourier basis with a modified-Chebyshev parameterization while holding all other conditions identical produces a qualitative reversal: $f_2 \leq 6.2\%$ at all three Reynolds numbers, while $f_t > 93\%$ (Table **??**, Figure **??**). This qualitative reversal upon changing only the parameterization basis demonstrates that modal concentration is conditional on the chosen control representation, not a universal property of the controlled cavity flow. The statement is about the controls obtained in these runs, not about the bases as such: within the same Fourier parameterization the representative at $Re = 1000$ is already markedly unsteady ($f_t = 32.2\%$), and the B_2 and B_3 branch realizations of Sec. **??** are time-dependent in that basis as well. A Fourier expansion therefore does not guarantee a quasi-stationary control, and the reversal reported here is a difference in the outcomes the optimizer produces, not a property of the basis algebra. The Chebyshev controls also overshoot the prescribed energy budget, with $E_r \approx 0.285$–0.291 against a target $E^* = 0.25$ (about 15% overshoot, versus 0.254–0.269 for the Fourier basis); this is a constraint-satisfaction difference, not evidence of higher dissipation. Basis construction details are provided in Supplementary Sec. S7.

INDEPENDENT MRT–LBM VERIFICATION AND GRID UNCERTAINTY

MRT-LBM solver and benchmark verification

An independent multiple-relaxation-time D2Q9 lattice-Boltzmann (MRT-LBM) solver was used to evaluate selected PINN-identified controls in a completely independent computational environment. The solver employs the D2Q9 lattice with the MRT collision rule

$$\mathbf{m}^* = \mathbf{m} - \mathbf{S}(\mathbf{m} - \mathbf{m}^{\mathrm{eq}}), \quad \mathbf{f} = \mathbf{M}^{-1}\mathbf{m}^*, \tag{17}$$

where $\mathbf{m} = \mathbf{M}\mathbf{f}$ are moment-space populations, $\mathbf{S} = \mathrm{diag}(s_0, \ldots, s_8)$ is the diagonal relaxation matrix, and $\mathbf{m}^{\mathrm{eq}}$ are the equilibrium moments. Lid boundary conditions are imposed using the Zou–He velocity bounce-back scheme . The LBM implementation is entirely independent of the PINN framework: it uses a custom C++/OpenMP solver, a different spatial discretization (Cartesian lattice vs. scattered collocation), and a fundamentally different numerical method. Complete solver specifications are provided in Supplementary Sec. S8.

The solver was validated against the benchmark cavity solutions of Ghia et al. . Centerline velocity errors are reported in Table **??**. At $Re = 100$ the L_2 error in u is 0.00506 and the L_∞ error is 0.01283. At $Re = 1000$ the errors increase moderately to $L_2(u) = 0.01411$ and $L_\infty(u) = 0.03003$, consistent with finer velocity gradient structure and the 256×256 lattice used for benchmark comparison. These errors are within typical MRT-LBM accuracy at this resolution and confirm solver reliability for the controlled-flow study.

Controlled-flow dissipation comparison

Four controlled configurations were evaluated at $Re = 500$ and $Re = 1000$: (i) uniform lid ($U_{\mathrm{lid}} = 1$, the baseline); (ii) first-harmonic $\sin(\pi x)$; (iii) stationary second-harmonic $A_2 \sin(2\pi x)$ representing the B_1 branch; and (iv) the mean PINN control evaluated at $t = 0$, representing the ensemble-averaged physics-informed solution. The LBM verification uses pure analytical controls (and the mean PINN profile) rather than the full time-varying PINN output; the B_1 branch admits a mean time-dependent fraction of about 7.4%, so the comparison quantifies the approximation by the stationary second-harmonic form. The magnitude of that approximation is bounded in Supplementary Sec. S8.5: the full time-dependent Fourier control has a cycle-averaged dissipation of 1.22×10^{-2} at $Re = 500$, about 2.5% above the mean-field value of the stationary PINN profile, so the stationary transfer reproduces the temporal control to within a few per cent. Integrated viscous dissipation ε and percentage reduction versus uniform lid are reported in Table **??**. The reductions of 76.5% at $Re = 500$ and 82.3% at $Re = 1000$ are obtained at the same lid-velocity

scale as the uniform lid but *without matching the realized actuation energy* E_r of the different profiles. The matched-energy comparison of Sec. **??** gives the physically relevant ranking.

Both the stationary second-harmonic control $A_2 \sin(2\pi x)$ and the PINN mean control achieve reductions of approximately 73–83% in integrated viscous dissipation relative to the uniform-lid baseline at the same lid-velocity scale, confirming that these controls produce physical effects reproducible in an independent solver. The close dissipation values of $A_2 \sin(2\pi x)$ and the PINN mean control are particularly significant: the difference at $Re = 500$ is approximately 3.3% relative to the uniform baseline, indicating that the low-dimensional second-harmonic structure captures the principal character of the learned PINN control. These reductions, however, are not obtained at matched actuation energy E_r. The subsections below repeat the comparison at a common E_r and quantify its grid uncertainty; the outcome is that the B_1 branch is not an iso-energetic optimum.

Iso-energetic comparison of the stationary controls

The comparison above holds the lid-velocity scale fixed but not the actuation energy: E_r ranges from 0.17 to 1.0. To separate the effect of the *shape* of the lid profile from the effect of the *amount* of actuation, the four stationary controls were re-run with the lid amplitude rescaled to the common actuation energy $E^* = \int_0^1 U_{\mathrm{lid}}^2 \, dx = 0.25$: the uniform lid then uses $U_{\mathrm{lid}} = 0.5$, and the harmonic controls use the amplitude $\sqrt{2E^*} = 0.7071$ (the B_1 PINN mean profile is used as computed, with $E_r = 0.2503$, i.e. within 0.15% of the target). Rescaling is not applied analytically to the results of Table **??**, because the effective Reynolds number of the cavity depends on the lid amplitude and ε does not scale quadratically with E_r; all matched-energy values below are independent LBM runs. In those runs the mobility is fixed by the nominal Reynolds number at unit lid amplitude, so a control whose peak lid velocity is a realizes an effective Reynolds number $a\,Re$: at $E^* = 0.25$ the uniform lid is a $0.50\,Re$ cavity while the harmonic controls are $0.71\,Re$ ones. The iso-energetic ranking therefore compares shapes at equal actuation energy and not at equal effective Reynolds number, and this amplitude difference is part of what separates it from the iso-velocity comparison. Results are reported in Table **??**.

The ranking at fixed energy differs qualitatively from the ranking at fixed lid velocity. At matched actuation energy the first harmonic is the only stationary control that lowers the dissipation, while both second-harmonic controls raise it: on the finest grid the ratios to the uniform lid of the problem statement are 0.647 for $k = 1$, 1.076 for $k = 2$ and 1.134 for the B_1 mean profile, i.e. −35.3%, +7.6% and +13.4%. That lid carries a tangential-velocity discontinuity at the two upper corners and its own dissipation does not converge (Sec. **??**); the ratios built on it therefore inherit that drift (18–21% over the triple), and we do not treat them as converged estimates of the shape effect. To separate the shape comparison from the corner artifact, the same three controls were also normalized by a corner-regularized uniform lid at the same energy (cosine taper $\delta = 0.05$, Sec. **??**): the $N = 512$ ratios are then 0.823, 1.368 and 1.443, i.e. −17.7%, +36.8% and +44.3%, with Richardson estimates 0.802, 1.347 and 1.427 and fine-grid grid-convergence indices (GCIs) of 3.1%, 1.9% and 1.4%.

Two statements survive both normalizations, and they are what we retain. First, the sign of the ordering $k = 1 <$ uniform $< k = 2$ is reproduced at all three resolutions and under both references: the first harmonic lowers the dissipation and the second-harmonic controls raise it, which reverses the same-velocity ordering, where both sine controls are less dissipative than the uniform lid and $k = 2$ is the best of the three. Second, the magnitude is reference-dependent: normalizing by the sharp lid overstates the first-harmonic advantage (−35.3% against −17.7%) and understates the second-harmonic penalty (+7.6% against +36.8%), because the non-converged corner contribution inflates the denominator. The regularized-lid ratios are the better-defined estimates of the shape effect; the sharp-lid ratios are those directly comparable with the uniform baseline of Table **??**, and both are reported. The separation between $k = 2$ and the B_1 mean profile (1.368 against 1.443, about 5%) exceeds their GCIs (1.9% and 1.4%) but not by a wide margin, so we write $k = 2 \approx B_1$ rather than a strict ordering between them.

The 73–82% reductions of Table **??** therefore do not establish that the selected B_1 branch is an

energetically efficient control: they reflect the fact that the learned profile has a smaller amplitude than the uniform lid, and once this amplitude difference is removed the second-harmonic structure is a net *penalty* on the dissipation objective. This is the quantitative basis for the statement that B_1 is selected by the regularization and by the optimization path rather than by physical optimality.

Grid-convergence uncertainty analysis

The same three grids, $N \in \{128, 256, 512\}$, are used for all four controls at matched energy. The observed order of convergence p_{obs} and the fine-grid grid-convergence index are computed by Richardson extrapolation with safety factor $F_s = 1.25$, and are reported in Table **??**. The estimators are also applied to the ratios, because the reduction percentages quoted above are ratio quantities and their errors need not cancel. A triple is flagged, and its extrapolated value is not used, when it is non-monotone, when p_{obs} lies outside $[0.1, 4]$, or when the GCI exceeds 100%: in these cases the smooth-error assumption underlying the estimator is not credible.

The three controls other than the uniform lid converge: their absolute triples give $p_{\mathrm{obs}} = 1.09$, 1.04 and 1.01 with fine-grid GCIs of 3.9%, 5.6% and 6.0% ($k = 1$, $k = 2$, B_1 mean). The uniform lid does not. Its increments are 1.745×10^{-3} and 1.540×10^{-3} between successive grids, so they decay by only 12% per refinement ($p_{\mathrm{obs}} = 0.18$, GCI = 131%); the formal Richardson limit would be 2.25×10^{-2}, more than twice the finest-grid value, which is not a credible estimate of the continuous-limit dissipation and is not used anywhere in this paper. The behaviour is the signature of the corner singularity, which the sine controls do not have because their lid velocity vanishes at both corners. To confirm that attribution we repeated the entire study with a uniform lid carrying a cosine taper of width $\delta = 0.05$ at each upper corner, at the same energy $E_r = 0.25$ (Table **??**). The tapered lid is continuous, and it converges: $p_{\mathrm{obs}} = 0.96$, GCI = 7.8%, with an extrapolated $\varepsilon_\infty = 9.20 \times 10^{-3}$. The taper removes 21% of the dissipation at $N = 512$ and, more importantly, turns a 16% growth of ε between the two finest grids into 6%, i.e. the expected $p \approx 1$ rate. The discontinuity is therefore the cause of the non-convergence, not a property of the uniform profile as such.

The tapered lid is a different actuation profile, and it does not replace the discontinuous lid specified in the problem; it enters only as a diagnostic and as a converged reference for the ratio. Normalizing the three other controls by it gives the second block of Table **??**: all three ratio triples are monotone and converge with $p_{\mathrm{obs}} \in [0.82, 1.03]$ and GCIs of 1.4–3.1%, against a drift of 18–21% and two flagged triples when the sharp lid is used as reference. The sign of the ordering is the same under both normalizations, and is observed at each of the three grids; the magnitude is not.

Figure **??** shows the three readings: the absolute dissipation, the ratio to the sharp lid and the ratio to the regularized lid. The iso-energetic conclusion does not rest on the internal error estimate of a single campaign, which the corner singularity makes unusable for the sharp-lid ratios. It rests on three statements. First, the ordering is observed at the three resolutions tested and under both reference lids. Second, with the regularized reference it carries a GCI of at most 3.1%. Third, the same matched-energy problem was solved in an earlier, independent MRT-LBM campaign with a different iteration budget; that campaign reached optimization residuals of order 10^{-10} where the present runs stop at the 2×10^5 iteration cap with residuals of order 10^{-6} to 10^{-5}, yet the two agree to 0.7% on every dissipation value and to 0.3% on the $N = 512$ ratios (0.6466 against 0.6463 for $k = 1$, 1.0755 against 1.0721 for $k = 2$; the comparison is tabulated in Supplementary Sec. S8.4). We therefore quote the regularized-lid ratios as the converged estimate of the shape effect, the sharp-lid ratios as the values comparable with the baseline of Table **??**, and the inter-campaign spread as the reproducibility error, rather than a single-grid error estimate. Supplementary Sec. S8.4 gives the full convergence data, and Sec. S8.5 reports the cycle-averaged response of the temporal controls.

Quasi-stationarity of controlled flows

The temporal stability of controlled LBM flows is characterized by the kinetic-energy fluctuation ratio $R_K = K_{\mathrm{fluct}}/K_{\mathrm{mean}}$ (Table **??**, Figure **??**). The transferred Fourier B_1 series yields

$R_K \leq 0.65\%$ across all tested Reynolds numbers, well below the 1% quasi-stationarity threshold, confirming that the predominantly stationary second-harmonic Fourier control establishes a quasi-steady flow state. By contrast, the modified-Chebyshev series produces profoundly unsteady responses with $R_K \geq 44\%$ at all Reynolds numbers and $R_K = 847\%$ at $Re = 100$. The two representations therefore generate qualitatively different physical flow states: Fourier B_1 controls produce quasi-steady cavity flows, while Chebyshev controls produce strongly oscillatory flows. The oscillation is not a benign regime difference: it is a dissipation penalty. Averaged over the cycle, the full Chebyshev control dissipates $\bar{\varepsilon} = 4.06 \times 10^{-2}$ at $Re = 500$ (Table **??**), a factor of 3.4 above the most dissipative and 5.9 above the least dissipative stationary control of Table **??** at the same lattice resolution. Its time-mean field dissipates only $\varepsilon(\langle u \rangle) = 5.97 \times 10^{-4}$, an order of magnitude *below* the same controls; the difference is the dissipation carried by the fluctuations, and it shows that a mean-field estimate can invert the ranking of a strongly unsteady flow (Jensen's inequality). Evaluated on the cycle average, the quasi-stationary controls of the second-harmonic and full-B_1 Fourier series are therefore the low-dissipation states among the transfers compared here. The quasi-stationarity diagnostic thereby provides independent physical confirmation that the Fourier B_1 branch corresponds to a physically realizable steady-state control target.

Table **??** makes the distinction quantitative. For a periodic response the physically relevant index is the cycle average $\bar{\varepsilon} = T^{-1} \int_0^T \varepsilon(t)\, dt$, not the dissipation of the time-mean field $\varepsilon(\langle u \rangle)$, and Jensen's inequality gives $\bar{\varepsilon} \geq \varepsilon(\langle u \rangle)$ with equality only in the steady limit. The two columns of the table show how large the gap can become: it is 3% for the nearly steady Fourier control at $Re = 100$, but a factor 622 for the Chebyshev control, whose mean field retains almost no motion while its fluctuations carry the dissipation. The cycle average is evaluated from the eight phase-averaged velocity fields stored by the solver. For the quasi-steady Fourier control the response locks on the forcing period and the eight-point quadrature agrees with a direct per-step accumulation at $Re = 500$ to 0.1%; for the strongly unsteady modified-Chebyshev control, whose fluctuations carry 44% of the kinetic energy, the same comparison shows the eight-phase value to be 5% low, and the direct accumulation is used instead (Supplementary Sec. S8.5).

The branch representatives can be placed on the same iso-energetic scale. The lid series of individual realizations of the $Re = 500$, $E^* = 0.25$ cell of the campaign map were rescaled by a single multiplicative factor so that the *cycle-mean* actuation energy is exactly E^*. Because the factor is uniform in space and time it leaves the shape observables f_2, f_t and C_3, and therefore the branch label, unchanged; only the amplitude, hence the effective Reynolds number, is affected (a series scaled by a factor s realizes an $s\,Re$ cavity at fixed mobility, as above). Each series was then advanced with the direct per-step accumulation above, stopping when the drift between successive hundred-period windows falls below 2×10^{-3}. The mean-field dissipation of the B_2 seed-5 and B_3 seed-0 transfers reproduces the earlier independent campaign to 0.01%, so the two solvers agree on the same input series. Table **??** gives the result.

Three of the branch transfers are unsteady enough that $\bar{\varepsilon}$ and $\varepsilon(\langle u \rangle)$ no longer describe the same flow: the B_2 realizations dissipate 3.1×10^{-5} and 1.8×10^{-4} in their mean field but 3.76×10^{-2} and 3.70×10^{-2} over the cycle, a factor of 1214 and 203, and for the first of them the mean field holds only 6% of the cycle-mean kinetic energy (1.55×10^{-5} against 2.47×10^{-4}). On the cycle average, B_2 is therefore *not* a low-dissipation branch: at matched energy it costs a factor of 5.5 above the least dissipative stationary control of Table **??** and comes within 7% of the modified-Chebyshev series, and the within-branch spread between its two seeds is only 1.7%. The two B_3 realizations fall in between (2.18×10^{-2} and 2.69×10^{-2}, a within-branch spread of 23%), i.e. below B_2, so the order in which the branches become accessible is not the order in which they perform. Combined with the B_1 series itself, which costs 1.22×10^{-2} on the cycle average against 1.17×10^{-2} for its own time-mean field (a 4% difference, the smallest of the three branch families), this closes the branch comparison at matched energy: the classification of Sec. **??** describes what the optimizer finds, not what the flow would prefer. The field-level counterpart of this comparison is Figs. **??**–**??** and Table **??**.

Figure **??** places the same numbers on a logarithmic scale, where the two regimes separate.

The five quasi-stationary controls span less than a factor of two, from 6.88×10^{-3} for $k = 1$ to 1.22×10^{-2} for the full B_1 series, and their mean-field and cycle-mean values coincide to within 4%. The four branch transfers with substantial temporal content leave that band: on the cycle average they cost 2.18×10^{-2} to 3.76×10^{-2}, i.e. 2.3 to 4.0 times the uniform lid at the same actuation energy, while their mean-field dissipation falls below every stationary control by more than two orders of magnitude in the B_2 case. The vertical extent of each segment in the left panel is therefore itself a diagnostic: it measures how much of the reported dissipation is carried by the unsteady part of the response rather than by its mean, and it is the quantity that a stationary comparison cannot see.

DISCUSSION: CONDITIONAL BRANCH SELECTION

What is actually selected by the physics-informed optimization?

The experiments presented above establish that the output of the physics-informed optimization is not a deterministic map $(Re, E^*) \to \mathcal{B}$ but rather an empirical frequency distribution over accessible branch classes that depends on the full set of variational parameters: $\mathcal{B}(Re, E^*, \lambda_{\text{var}}, \text{seed}, \text{capacity}, \text{basis}, AR)$. This finding has a direct methodological implication: a single converged solution can only be said to be a solution that the adopted optimization procedure can reach; it cannot, without additional multi-start evidence, be said to be the unique or globally optimal solution of the underlying control problem.

The co-existence of B_1 and B_2 realizations with total losses within a 1.7% spread is the most direct evidence for competing outcome clusters under the adopted formulation. If one branch were substantially more optimal in terms of the weighted training loss, it would achieve measurably lower losses, which is not observed. The near-equal global empirical recovery frequencies of B_1 (42.2%) and B_2 (43.8%) further confirm that neither branch dominates globally. The concept of a "dominant" mode must therefore be replaced by "frequently accessible under the given formulation." Similar total losses do not imply comparable physical objectives, and loss decomposition must be used for direct physical comparison. Table S9 gives that decomposition for one representative of each class: the three totals agree to 0.3% (1.3741, 1.3725, 1.3768), whereas the dissipation term varies by a factor 2.2 (2.376, 5.149, 2.491×10^{-4}). All three representatives sit at the same realized energy, $E_r = 0.2595$, with an energy-constraint residual $(E_r - E^*)^2$ of 9.0–9.1 $\times 10^{-5}$, i.e. $|E_r - E^*| = 9.5 \times 10^{-3}$ or 3.8% of E^*. The energy budget is therefore common to the three classes and the dissipation term is the one that differs. The LBM ranking of Table **??** is likewise formed at a common realized energy, so the physical comparison is not biased by a different actuation budget.

The residual class B_3, accounting for 14.1% of realizations, is heterogeneous rather than a single outcome family: 15 of its 26 members are dominated by temporal content, 8 by the stationary second harmonic and 3 by harmonics of order $m \geq 3$ (Table **??**). It is selectively favored by high actuation energy and, in the ablation campaign, by intermediate variance regularization. Removing the variance term entirely does not produce B_3 at all: it produces first-harmonic controls, i.e. the B_{first} class of Sec. **??**. Its less frequent appearance in the reference campaign suggests it is less frequently recovered under standard loss weights, but the $\lambda_{\text{var}} = 0$ results show that flattening the loss landscape in the spatial-variance dimension replaces the selected family outright rather than merely reshuffling B_1 against B_2.

Figures **??**–**??** and Table **??** show what the three labels mean at the level of the fields. The three captures use one protocol: the same solver, the same $N = 256$ lattice, the same $Re = 500$ and the same realised actuation energy (0.2500, 0.2504 and 0.2503 in units of E^*), two hundred forced periods of settling followed by a drift test on $\varepsilon(t)$ and sixty-four forced periods of sampling, and no rescaling of the lid amplitude, so the control that enters the solver is the one the PINN produced rather than an idealisation of it. All three settling phases ended at the five-hundred-period cap, with a relative drift over the last window of 1.5×10^{-3}, 2.6×10^{-9} and 7.3×10^{-5}.

The time-mean fields already separate the classes. The B_1 flow is the familiar pair of counter-rotating cells, centred at $(0.31, 0.54)$ and $(0.69, 0.56)$, with $\max|\langle \mathbf{u} \rangle| = 0.733$ and $\langle \psi \rangle \in [-0.0362, 0.0354]$; half of its mean kinetic energy lies below $y/L = 0.675$, so the cells fill the cavity.

The instantaneous states coincide with that mean to 2.7% in the field norm, and $R_K = 4 \times 10^{-4}$, so the four instantaneous panels of the other two rows are replaced in row 1 of Fig. **??** by the quantitative statement; the fields themselves are in Supplementary Fig. S5. The B_2 mean field is 19 times weaker in stream function ($\max|\langle\psi\rangle| = 0.0019$) and 18 times weaker in peak speed (0.042, i.e. 5.7% of the B_1 value); its three mean extrema all sit at $y/L = 0.89$–0.92, and half of the mean kinetic energy is above $y/L = 0.95$, so the mean motion is a shallow recirculation under the lid, not a cavity-filling cell. This is what a control with $f_t = 0.998$ leaves behind: the oscillation is 97% of the instantaneous field, the instantaneous circulation is one to three times the mean one, and the mean is small because the lid profile reverses sign twice per period, not because the flow is weak. The B_3 flow keeps a mean circulation of intermediate strength ($\langle\psi\rangle \in [-0.0305, 0.0219]$, $\max|\langle\mathbf{u}\rangle| = 0.595$, 81% of the B_1 value) but places it differently: its two extrema sit at $(0.20, 0.86)$ and $(0.77, 0.81)$, close to the lid, and half of the mean kinetic energy is above $y/L = 0.91$. The label B_3 denotes the absence of a dominant spectral family rather than a blend of unrelated harmonics. For the representative used here the first spatial harmonic carries $f_1 = 0.0063$, against $f_2 = 0.520$ and $f_t = 0.448$: the control balances the stationary second harmonic against the temporal harmonic with essentially no first-harmonic content, so it satisfies the "neither dominance" definition without carrying an unrelated first-harmonic component.

Figure **??** puts the same fields on scales common to the three columns—one colour range per row, one normalisation U_{ref}, no per-panel rescaling—and writes the occupied range in each panel, so that a reader can check the common scale against the field. The mean field falls from ± 0.73 for B_1 to 0.042 for B_2 and 0.59 for B_3, while the streamwise fluctuation field does the opposite: $u'_{x,\mathrm{rms}}$ peaks at 0.073 for B_1, and even there only in the lid layer, since its interior maximum is 0.0036, i.e. 5% of the peak. The B_2 fluctuation reaches 0.555, thirteen times its own mean and 76% of the B_1 mean peak, and the B_3 one reaches 0.416. "Weak mean flow" is therefore not "weak flow", and a figure that normalised each panel by its own maximum would hide exactly the contrast that defines the two unsteady branches. The centreline profiles of panels (d) and (e) make the same point in one dimension. Note that B_1 is not exactly stationary either: its temporal content ($f_t = 0.074$) produces a 7.3% lid-layer rms and $R_K = 0.0004$, so an identically zero fluctuation row is what a strictly stationary control would give, not what the full PINN series gives.

Figure **??** and Table **??** give the budget of the three captures, all four quantities being cycle averages over the sixty-four sampled instantaneous fields. The actuation energy realised by B_1 stays within 23% of the target over the period (0.192–0.286), that of B_3 within 35% (0.163–0.336), while B_2 swings between 0.4% and 236% of E^*: its lid is switched off and driven at twice the target in the same period, and the cycle average, not the instantaneous value, is the quantity common to the three branches. The kinetic energy separates them by two orders of magnitude (1.15×10^{-2}, 2.47×10^{-4} and 3.98×10^{-3}, i.e. 2.1% and 34% of the B_1 value) and is constant within each sampled window to 1.1×10^{-3} in relative terms (the B_3 value; the other two are an order of magnitude smaller), as expected for a locked periodic state. The cycle-mean dissipation orders them the other way round: 1.22×10^{-2} for B_1, 3.76×10^{-2} for B_2 and 2.18×10^{-2} for B_3, so at matched energy B_2 costs 3.1 times B_1 and 1.7 times B_3. Within the period, $D(t)$ varies by a factor 1.5, 10.9 and 4.2 respectively, and the lid power of B_2 is negative for 14% of the period, i.e. the flow returns energy through the lid while the Stokes layer spins down. The Jensen ratios of Table **??** make the consequence explicit: a diagnostic built on the time-mean field ranks B_2 as the least dissipative of the three by three orders of magnitude, whereas the cycle average ranks it as the most dissipative. The residual $\langle P_{\mathrm{lid}}\rangle - \langle D\rangle$ is -2.6%, $+5.6\%$ and $+3.8\%$, two to three orders of magnitude above the drift of K over the window; it is the truncation error of the first-order wall-shear estimate, not a physical imbalance, and the same estimator is used for the three branches. The block standard errors of $\bar{D}$ are below 0.1% because the response is periodic; the physically relevant uncertainty is the seed-to-seed spread of a class, 1.7% within B_2 and 23% within B_3, both reported with the iso-energetic table.

Physical origin of second-harmonic recurrence and iso-energetic performance

The stationary second-harmonic structure has a clear geometric rationale. The antisymmetric

profile $A_2 \sin(2\pi x)$ creates a two-lobe actuation pattern at the upper lid, generating a pair of counter-rotating recirculation cells rather than the single primary vortex of the uniform-lid cavity. This can produce a more symmetric velocity gradient distribution, but its physical desirability depends strongly on the energy basis of comparison.

Two diagnostics make that rationale quantitative. Figure 4 shows the steady response of the three controls at the matched actuation energy, on scales common to the three columns. The uniform lid concentrates its dissipation in the two upper corners, where the imposed tangential velocity jumps to zero: the peak strain-rate magnitude there is 169 in units of U_{ref}/L, against 15.0 for $k = 1$ and 23.4 for $k = 2$, and those corner layers are the brightest features of panel (b). Both sine controls remove the jump, and their dissipation is spread over the lid shear layer and the side walls instead. The two-lobe topology is visible in panel (a): the $\sin(2\pi x)$ lid produces a pair of counter-rotating cells of nearly equal strength separated by a wall-to-wall separatrix at $x = 0.5$, each enclosing its own low-strain core, whereas the uniform and $k = 1$ lids recover the classical single primary vortex with a corner eddy. The price of that reorganization appears in panel (c): the strain-rate density of the $k = 2$ response is depleted at low $|S|$ and shifted towards larger values, with an interior mean of 1.450 against 1.039 for the uniform lid and 1.088 for $k = 1$, i.e. a 40% larger mean strain at identical actuation energy. This is the local origin of the dissipation penalty of the second harmonic at matched energy: +37% against the corner-regularized uniform lid at $N = 512$, and +8% against the sharp lid, whose own dissipation is not converged (Table **??**): the second harmonic buys a symmetric two-cell state with a stronger interior strain field, and the corner singularity it removes is not costly enough to compensate. The trade-off is explicit: $k = 1$ removes almost all of the corner peak (169 to 15.0) at a cost of +4.7% in interior mean strain (1.088 against 1.039), whereas $k = 2$ removes slightly less of it (to 23.4) while raising the interior mean by 39.6%. The second harmonic therefore pays roughly eight times the interior penalty of the first for no additional corner gain. Integrating the plotted fields reproduces the tabulated dissipations to within 0.7% for the three controls, which also serves as an internal consistency check of the transfer.

Table **??** shows large dissipation reductions relative to the uniform lid at the same lid-velocity scale, but the realized actuation energies differ. The iso-energetic comparison of Table **??** reverses the ranking: at equal actuation energy the first-harmonic control $\sin(\pi x)$ is the least dissipative of the four stationary profiles, while the stationary second-harmonic control $A_2 \sin(2\pi x)$ and the B_1 PINN mean profile are both *more* dissipative than the uniform lid. The B_1 branch is therefore a regularization-selected cluster, not a low-dissipation mechanism at fixed energy: its advantage disappears as soon as the comparison is made at equal actuation energy rather than equal lid velocity.

The quasi-stationarity of the Fourier B_1 flow ($R_K \leq 0.65\%$) confirms that the selected profile generates a steady cavity state, while the Chebyshev controls sustain a persistent oscillation ($R_K \geq 44\%$) whose energetic cost is carried by the fluctuations: their time-mean field retains only a thin residual motion and its dissipation (5.97×10^{-4}) lies nearly two orders of magnitude below the cycle average (4.06×10^{-2} at $\bar{E}_r = 0.286$). Both the representation and the quasi-stationarity constraint therefore condition the outcome. Within the stationary Fourier class the first harmonic is the least dissipative member, and the modified-Chebyshev class does not reach a lower-dissipation regime once its cost is evaluated over the cycle. The iso-energetic ranking is accordingly restricted to the stationary controls of Table **??**, for which a single time-independent dissipation value is defined; the temporal branches are ranked on the cycle average at matched cycle-mean energy (Table **??**, Table **??** and Supplementary Sec. S8.5).

The persistence of the second-harmonic structure across the capacities and truncations tested shows that it is a robust feature of the variational landscape within those ranges, not that it is a physical optimum: a feature can be reproducible and still be disfavoured by the physical objective once the comparison is made at fixed actuation energy. The erosion of f_2 at high energy ($E^* = 2$) and at compressed geometry ($AR = 0.5$) further shows that the dominance of the second-harmonic outcome is not unconditional: when the energy budget or geometry changes sufficiently, other

modal strategies become competitive. The global balance between B_1 and B_2 accessibility in the broad campaign reflects this competitiveness between stationary and time-dependent dissipation strategies.

Optimization bias versus fluid mechanics

The loss-ablation results demonstrate that the dominant driver of B_1 selection is $\mathcal{L}_{\text{var}}$, not $\mathcal{L}_{\text{diss}}$. Removing $\mathcal{L}_{\text{diss}}$ leaves f_2 essentially unchanged (92.3% vs. baseline 92.2%), while removing $\mathcal{L}_{\text{var}}$ causes a complete modal-class shift to $f_2 = 0.2\%$. This means that in the reference loss formulation, the spatial-variance penalty acts as the primary optimization landscape shaper, directing the search into the cluster of the spatially structured B_1 control. The physical dissipation objective, while present and contributing to overall convergence quality, plays a secondary role in determining which outcome cluster is recovered. The iso-energetic LBM comparison shows that this selected branch is suboptimal at fixed actuation energy relative to the first harmonic, underscoring that regularization-driven selection does not equate to physical optimality.

This does not invalidate the realizability of the identified controls: the LBM verification confirms that the controls genuinely reduce viscous dissipation relative to a uniform lid of the same lid velocity, and the quasi-stationarity analysis confirms they are physically realizable. What the ablation result reveals is a distinction between *which loss term selects the branch* and *which physical property justifies the selected branch*. The variance regularization acts as an implicit basis selector, preferring spatially structured controls among competing optima; the dissipation objective then evaluates those controls, and at fixed actuation energy it does not rank the second-harmonic structure above the first harmonic or the uniform lid. Practitioners designing physics-informed control problems should be aware that the loss weight configuration may be as consequential as the physical objective in determining which branch is selected.

Scope, limitations, and outlook

Scope of the formulation. The results pertain to two-dimensional incompressible flow in a square cavity, to quasi-stationary control objectives, and to actuation with fixed spatial support on the upper lid. Two levels of evidence must be distinguished. The stationary ranking rests on four controls compared at a common realized energy on three grid resolutions and against two reference lids; within those resolutions and references it is complete. The temporal comparison is narrower: the three branch representatives are transferred without retraining and ranked at matched cycle-mean energy on a single grid, with two realizations per branch, and retraining under an explicit cycle-mean energy constraint has not yet been performed. The claims made in Secs. **??**–**??** are confined to these controls and to this geometry.

Numerical uncertainty. For the sharp uniform lid the absolute dissipation is not converged on the present grid triple, because the corner velocity discontinuity violates the smooth-error assumption of the grid-convergence estimator; the observed order is recovered once the corner is regularized, and the sign of the iso-energetic ordering is insensitive to that regularization. The temporal branches are ranked at matched cycle-mean energy on a single grid ($N = 256$, Table **??** and Supplementary Sec. S8.5), with two realizations per branch. Their wall-power budget closes to within 6% (Fig. **??**d), and the cycle means reproduce the published values to better than 0.04% for B_2 and B_3. A grid study of the temporal branches and retraining under an explicit cycle-mean energy constraint, instead of the post-hoc amplitude rescaling documented in Supplementary Sec. S8.6, are the natural next steps.

Statistical resolution. Individual cells of the (Re, E^*) campaign contain few realizations, so the recovery frequencies are reported as conditional rates with Wilson confidence intervals (Table **??**) rather than as fixed probabilities. The branch representatives at $E^* = 0.25$ are single realizations and are labelled as such wherever they are used. They are representative rather than extreme: the B_2 representative lies 0.021 from the centre of its class (median distance 0.014, class radius ≤ 0.028) and its $\bar{D}$ differs by +1.7% from the other seed; the B_3 representative is the

low-dissipation member of its class, the other seed being 23% more dissipative, so the B_3 numbers in Table **??** should be read as a lower bound on that class.

Interpretation. The branch realizations at $E^* = 0.25$ exist only at $Re = 500$; at $Re = 100$ and $Re = 1000$ no B_2 or B_3 realization was recovered at the target energy, so the iso-energetic branch comparison is restricted to $Re = 500$. The representation-dependence result covers three Reynolds numbers and a single modified-Chebyshev parameterization, and the symbolic expressions describe the shape of the mean profile, not its energy budget: the energy reconstructed from a fold expression differs from that of the data by -17% to $+10\%$. Finally, because the PINN imposes no initial condition, the branch comparison concerns periodic states rather than trajectories from a prescribed initial field.

Outlook. Three extensions follow directly. Three-dimensional cavity flow would introduce spanwise modes and possible Taylor–Görtler structures, and would test whether the second-harmonic state survives when the flow can vary along the span. An explicitly unsteady objective would probe branches that a quasi-stationary objective cannot reach. Finally, other basis families (Zernike polynomials, wavelets, data-driven POD modes) would establish how general the representation dependence is; that question requires multi-representation campaigns of the type conducted here rather than physical argument alone. The symbolic expressions developed in Sec. **??** provide a compact starting point for an analytical treatment of the control mechanism at larger Reynolds numbers.

SYMBOLIC REGRESSION OF THE MEAN CONTROL

Symbolic regression of the mean-control profile

The B_1 Fourier controls share a common spatial structure, and we ask whether that structure admits a compact algebraic description. PySR symbolic regression is applied to the time-mean lid profile of the B_1 representative of each Reynolds condition of the $E^* = 2.0$ slice, peak-normalized, with inputs (x, Re) and a single output, the normalized profile $\bar{U}_{\mathrm{lid}}(x; Re)/\max_x |\bar{U}_{\mathrm{lid}}|$. The search uses binary operators $\{+, -, \times, /\}$, the unary operators sin, cos, exp, `square` and $\sqrt{}$, an $L2$ distance loss, twenty populations of one hundred iterations and a maximum expression size of thirty, and returns the best complexity-accuracy trade-off on the Pareto frontier. The dataset and search specifications are provided in Supplementary Sec. S9.

Two properties of the recorded output constrain how the result can be reported. First, the regression target is a cycle-mean profile, so no time variable enters any of the returned expressions: the description is a function of (x, Re) only. Second, PySR is a stochastic symbolic search rather than a parametric fit with fixed functional form, so the expression depends on the fit that produced it. The six leave-one-condition-out refits of Sec. **??** returned six structurally different members of the Pareto frontier, and no closed form is common to all of them. The reproducible content is therefore the set of recorded expressions, listed in Supplementary Sec. S9, together with their transfer scores and their errors, rather than a single formula. The expression quoted below is the fit held out at $Re = 1000$, reproduced exactly as recorded:

$$\begin{aligned} U_{\mathrm{SR}}(x; Re) = {} & -1.3919\,x \\ & + 0.6467\,\sin[(0.0003995\,Re - 2.7101)(3.4468\,x - 1.7234) + 0.3135] \\ & + 0.7839, \end{aligned} \tag{18}$$

to be read on $x \in [0, 1]$ and combined with the peak scale of its condition (2.148 at $Re = 1000$) to restore the dimensional profile. The nested-sine argument $3.4468\,x - 1.7234$ reappears in three of the six recorded fits, which is the sense in which the B_1 family shares a spatial frequency. Its wavelength on the unit cavity is $2\pi/3.4468 = 1.82$ and its phase shift is exactly half of it ($1.7234 = 3.4468/2$), so the sine vanishes at the cavity centre $x = 1/2$ and is antisymmetric about that point; taken with the linear term, this is what reproduces the two-lobe velocity pattern of the branch. No frequency, phase or amplitude in Eq. ? is a fitted parameter of a prescribed form: they

are the constants returned by the search. The expression is a shape descriptor, not an admissible control: it does not enforce the corner condition $U_{\mathrm{SR}}(0) = U_{\mathrm{SR}}(1) = 0$, nor the actuation-energy constraint $E_r \approx E^*$, and rescaled and integrated it reproduces the energy of the data profile only to within -17% to $+10\%$ (Sec. **??**); it is used here to summarize the spatial structure of the branch, not as a candidate lid law.

Leave-one-Reynolds-condition-out validation

A leave-one-Reynolds-condition-out (LOCO) cross-validation is performed across the six held-out Reynolds values $Re \in \{100, 300, 500, 700, 900, 1000\}$ at which B_1 realizations are populated in the campaign; the intermediate values $Re \in \{400, 600, 800\}$ contain no B_1 realization and are not held out. For each held-out condition, PySR is re-fit on the remaining five conditions, and the expression that fit returns is evaluated on the held-out data. The six fits are therefore six different expressions, and the scores below are the transfer scores of those six expressions, not the score of one formula tested six times; each expression is listed in Supplementary Sec. S9 with its own metrics. Figure **??** shows the LOCO R^2 scores. They range from $R^2 = 0.8238$ (worst case, $Re = 100$) to $R^2 = 0.9718$ (best case, $Re = 500$), with a mean of 0.914. The corresponding root-mean-square errors on the peak-normalized profile lie between 0.113 and 0.252, with a mean of 0.18: the shape is reproduced to under a fifth of the profile peak. What the transfer does *not* preserve is the amplitude: when each fold expression is rescaled by the peak scale of its condition and integrated, the actuation energy of the reconstructed mean profile differs from that of the data profile by -17% to $+10\%$ across the six folds, as tabulated in Supplementary Sec. S9. This demonstrates that the symbolic expressions capture a genuine cross-condition structure of the B_1 control family, with the $Re = 100$ fold remaining the most demanding extrapolation, while confirming that they describe the shape of the control rather than its budget.

Scope and interpretation of the symbolic result

The recorded expressions establish two things and cannot establish a third. They show that the B_1 family is smoothly describable: every fold transfers with $R^2 \geq 0.82$, three of the six return the same nested-sine argument, and the profiles therefore behave as members of one family varying smoothly with Re. They also give a compact, training-free starting profile for future studies, to be used with the peak rescaling above. They do not constitute a reusable control law: the constants are those of single runs of a stochastic search, no closed form is common to the folds, and the reconstruction error on the actuation energy, up to 17% of the condition budget, is of the same order as the energetic differences a control comparison is asked to resolve. Read with these limits, the symbolic result is a compact description of the recovered family, not an independent proof of optimality. High LOCO R^2 values establish that the branch structure is internally consistent, not that the branch is globally optimal.

CONCLUSIONS

This study has quantified how a low-dimensional wall actuation selects among competing states of the lid-driven cavity, and what the selected state costs in dissipation. The principal findings are as follows.

1. **Three empirical branch clusters are reproducibly identified.** A broad (Re, E^*) campaign of 185 independent physics-informed neural network (PINN) optimizations reveals three empirically distinguishable clusters of converged outcomes: B_1 (stationary second-harmonic dominated, 42.2%), B_2 (temporally dominated, 43.8%), and B_3 (14.1%), a residual class in which neither a stationary harmonic nor the explicit time dependence dominates. A full-spectrum audit adds a fourth, first-harmonic class B_{first} to the classification rule; it is empty in this campaign ($f_1 \leq 12.2\%$) but is the actual outcome of the $\lambda_{\mathrm{var}} = 0$ ablation. The same audit shows that B_3 is not a homogeneous family: 15 of its 26 members are temporal-dominated, 8 second-harmonic-dominated and 3 higher-harmonic-dominated. The near balance of B_1 and B_2 shows that neither branch uniquely dominates the optimization landscape; both are frequently accessible under the adopted formulation. The numerical experiments behind these numbers are itemized in Table S22, which lists each campaign

with its method and run count; several studies re-use the same trained controls, so no deduplicated grand total is claimed.

2. **Branch accessibility depends on the optimization formulation.** The variance-regularization weight λ_{var} is the dominant determinant of branch selection. At $\lambda_{\text{var}} = 0$, all conditions yield first-harmonic controls (B_{first}), not the residual B_3 class; increasing λ_{var} non-monotonically shifts accessibility between B_1 and B_2 in a way that depends on the operating point. The branch structure is jointly a property of the fluid problem and the variational formulation, not of the fluid problem alone.

3. **Random initialization produces competing branches of comparable loss quality.** Ten independent seed experiments at $Re = 500$, $E^* = 0.25$ produce 4 B_1, 4 B_2, and 2 B_3 realizations with total training losses clustered within $[1.371, 1.394]$, a relative spread of only 1.7%. The competing clusters correspond to outcomes with comparable training losses under the weighted objective, not to a unique global minimum. Similar total losses do not imply comparable physical performance; loss decomposition shows different contributions across branches.

4. **The stationary second-harmonic branch is robust to network capacity and control-space dimension.** The B_1 branch persists across all tested network capacities (4×32 to 8×128) and Fourier basis truncations (6 to 72 nominal modes), with $f_2 \in [87.3\%, 92.3\%]$. This robustness, over the capacity and truncation ranges tested, shows that the structure is not a numerical artifact of insufficient expressivity.

5. **Modal concentration is representation-conditional.** For the three Reynolds numbers and the single modified-Chebyshev parameterization tested, replacing the Fourier basis reverses the modal outcome: the Chebyshev controls are exclusively time-dependent ($f_t > 93\%$) and produce highly unsteady cavity flows (ratio of fluctuating to mean kinetic energy $R_K \geq 44\%$), while the Fourier B_1 controls establish quasi-stationary flows ($R_K \leq 0.65\%$). The same Fourier basis yields unsteady B_2 and B_3 realizations, so the second-harmonic branch B_1 is specific to the Fourier parameterization tested rather than to the cavity problem at large.

6. **Iso-energetic ranking reverses the velocity-matched ranking.** Transfer of the $A_2 \sin(2\pi x)$ and PINN mean controls to an independent multiple-relaxation-time lattice-Boltzmann (MRT-LBM) solver gives reductions of $\sim 76.5\%$ at $Re = 500$ and $\sim 82.3\%$ at $Re = 1000$ in integrated viscous dissipation relative to the uniform-lid baseline *at the same lid-velocity scale*. Iso-energetic comparisons within the stationary class, at the common energy $E^* = 0.25$, give the reverse order $k = 1 <$ uniform $< k = 2$ (B1) at each of the three grid resolutions tested. The sign of this ordering survives normalization by a corner-regularized uniform lid, against which the ratio triples converge with grid-convergence indices (GCIs) of at most 3.1%; the magnitude is reference-dependent (between -35% and -18% for $k = 1$, and between $+8\%$ and $+37\%$ for $k = 2$, depending on the reference). The frequently selected B1 branch is therefore regularization-selected and suboptimal at fixed energy. The strongly unsteady modified-Chebyshev control, by contrast, is *more* dissipative once its cost is averaged over the oscillation cycle ($\bar{\varepsilon} = 4.06 \times 10^{-2}$ at $Re = 500$, against 1.22×10^{-2} for the full Fourier control at $\bar{E}_r = 0.250$); the low value obtained from its time-mean field is an artifact of time-averaging. The unsteady alternative therefore does not improve on the stationary Fourier family. The close dissipation values of the analytic second-harmonic and the PINN mean control show that the low-dimensional structure captures the principal character of the learned control.

7. **Symbolic regression provides a compact, generalizing description of the recovered family.** Each leave-one-Reynolds-condition-out PySR refit of the peak-normalized mean

B_1 profile transfers to the held-out condition with $R^2 \in [0.8238, 0.9718]$ (mean 0.914) across the six Reynolds conditions populated by B_1 realizations; the nested-sine argument $3.4468\,x - 1.7234$ is returned by three of the six independent fits. The positive LOCO scores confirm that the B_1 branch forms a coherent family of related controls varying smoothly and predictably with Reynolds number over the range where it is reliably recovered. They describe a shape, not a budget: the actuation energy reconstructed from a fold expression differs from that of the data profile by -17% to $+10\%$, so the expressions are a compact description rather than a reusable control law.

The findings reduce to a hierarchy of properties:

$$\boxed{\text{branch existence} \neq \text{branch accessibility} \neq \text{branch reproducibility} \neq \text{global optimality.}} \quad (19)$$

The second-harmonic B_1 branch has the first three: it exists as a physically realizable flow state, is accessible under the variational formulations tested, and reproduces across computational platforms. It does not have the fourth. At fixed actuation energy it dissipates more than the uniform lid and more than the first harmonic. The weighted training losses of the B_2 and B_3 states are nearly indistinguishable from that of B_1, but this equivalence is confined to the optimization objective: the cycle-mean dissipations of the three states differ by up to a factor of three. B_1 is therefore neither a unique optimum of the training loss nor an energetic optimum at fixed energy.

The three branches differ physically in how the actuation energy is deposited. B_1 maintains a quasi-stationary two-cell circulation that fills the cavity ($\max|\mathbf{u}| = 0.73$, interior $u'_{\mathrm{rms}} = 0.0036$, $R_K = 4 \times 10^{-4}$). B_2 leaves only a shallow lid-layer mean ($\max|\mathbf{u}| = 0.042$) and puts its dynamics in the oscillation ($u'_{x,\mathrm{rms}} = 0.555$), returning energy through the lid for 14% of each period. B_3 displaces the two cells towards the lid (extrema at $y/L = 0.81$ and 0.86) and fluctuates at three quarters of the B_2 level. Cycle-mean dissipation separates these states more sharply than any mean-field diagnostic: at matched energy B_2 costs 3.1 times B_1 and 1.7 times B_3, an ordering that a diagnostic built on $\langle\mathbf{u}\rangle$ reverses by three orders of magnitude. That contrast is the physical content of the branch structure. The optimizer can reach stationary, strongly modulated, and displaced-cell states, and the state it reaches most often under strong variance regularization is not the cheapest at fixed energy.

The combination of multi-start PINN campaigns, systematic sensitivity analyses, and independent lattice-Boltzmann verification used here is directly transferable to other confined flows in which several controlled states compete.

Supplementary Material The Supplementary Material accompanying this article contains the implementation details, the complete campaign tables, the auxiliary figures, and the reproducibility material that support the analysis above. It is organized in ten sections and contains six figures (S1–S6) and twenty-two tables (S1–S22).

S1. Detailed PINN implementation: network architectures (Table S1), normalization and sampling rules, the optimization schedule, and the explicit form of every loss term (Sec. S1.4).

S2. Control-space and modal-analysis details: the Fourier and modified-Chebyshev bases, the definitions of the control-energy fractions f_1, f_2, f_t and of the concentration C_3, the branch-classification rule, and the spectral fingerprints.

S3. The 185-realization campaign, cell-aggregated: branch counts, modal composition, and energy calibration for every operating point of the (Re, E^*) grid (Tables S2 and S3, Fig. S1).

S4. Complete λ_{var} campaign and confidence intervals: the full sweep and the Wilson score intervals used for the recovery frequencies (Table S4).

S5. Energy, initialization, and loss-ablation studies: the energy sweep, the ten-seed metrics, the eight-configuration loss ablation, and its graphical representation (Fig. S2).

S6. Network-capacity and control-truncation studies.

S7. Geometry and control-parametrization studies: cavity aspect ratio and the modified-Chebyshev comparison (Fig. S3).

S8. Independent MRT–LBM verification and grid uncertainty: solver validation, the iso-energetic and corner studies, the grid-convergence analysis, the quasi-stationarity and cycle-mean validation, and the matched-protocol branch captures. This section contains the dissipation comparison (Fig. S4), the eight phase bins of the three branch representatives (Fig. S5), and the per-panel colour-scale version of Main Fig. 13 (Fig. S6).

S9. Symbolic-regression configuration and leave-one-Reynolds-condition-out details: PySR hyperparameters, per-fold metrics, and the six recorded expressions.

S10. Reproducibility and computational pipeline: software environment, the inventory of numerical experiments with their run counts (Table S22), and the scripts that regenerate every table and figure.

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Data Availability The data that support the findings of this study are available from the authors upon reasonable request.

We consider a two-dimensional incompressible flow in the square cavity $0 \le x \le 1$, $0 \le y \le 1$, made dimensionless with the cavity height L and a reference velocity U_0 (Fig. 1a). The lower and lateral walls are no-slip and stationary; the upper wall moves tangentially with a velocity profile $U_{\mathrm{lid}}(x,t)$ that is the control variable. The velocity field $\mathbf{u} = (u, v)$ and the pressure p satisfy

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u}\cdot\nabla)\mathbf{u} = -\nabla p + \frac{1}{Re}\nabla^2\mathbf{u}, \qquad \nabla\cdot\mathbf{u} = 0, \tag{20}$$

with $Re = U_0 L/\nu$, subject to

$$\mathbf{u} = \mathbf{0} \ \text{ on } x = 0,\ x = 1,\ y = 0, \qquad u = U_{\mathrm{lid}}(x,t), \quad v = 0 \ \text{ on } y = 1. \tag{21}$$

A pointwise tangential velocity along the lid idealizes an array of independently driven wall segments, or a distributed momentum source such as a dielectric-barrier-discharge actuator, whose near-wall forcing is equivalent in the time-averaged sense used here to a tangential wall velocity [13]. Because U_0 is a scaling velocity and not a property of the lid, a profile may change sign along x or in time; the amplitude of the actuation is fixed separately by the norm constraint introduced in Sec. 2.2.

2.2 The quadratic actuation norm and the mechanical wall power

Two different scalar quantities characterize a wall actuation, and they must not be conflated. The first is the *quadratic actuation norm*

$$E_{\mathrm{act}}(t) = \int_0^1 U_{\mathrm{lid}}^2(x,t)\,dx, \qquad \overline{E}_{\mathrm{act}} = \frac{1}{T}\int_0^T E_{\mathrm{act}}(t)\,dt, \tag{22}$$

which measures the squared amplitude of the commanded wall velocity, has units of $U_0^2 L$, and is the quantity held fixed in all comparisons below. It is a norm of the boundary data, not a work: it contains no information about the fluid response. The second is the *mechanical power* transmitted by the lid,

$$P_{\mathrm{lid}}(t) = \int_0^1 U_{\mathrm{lid}}(x,t)\,\tau_w(x,t)\,dx, \qquad \tau_w = \frac{1}{Re}\left.\frac{\partial u}{\partial y}\right|_{y=1}, \tag{23}$$

positive when the wall does work on the fluid. P_{lid} depends on the actuation and on the flow through the wall shear stress, and it is the quantity that enters the energy balance. Two actuations

of equal $\overline{E}_{\mathrm{act}}$ can therefore inject very different amounts of power; conversely, a large norm does not guarantee a large power if the wall shear is weak.

Holding $\overline{E}_{\mathrm{act}}$ fixed also fixes the root-mean-square wall velocity, $\overline{E}_{\mathrm{act}}^{1/2} = 0.5$ at the value $E^* = 0.25$ used throughout, but not the maximum wall velocity. A uniform lid of unit shape then has amplitude 0.5, whereas a first harmonic of the same norm has amplitude $\sqrt{2E^*} = 0.7071$; a peakier profile has a larger maximum still. Because the viscosity is fixed, a profile whose peak wall velocity is $U_{\max}$ realizes an effective Reynolds number $Re_{\max} = U_{\max} Re$ based on that peak. The comparisons below are therefore comparisons at equal quadratic actuation norm, at fixed viscosity and geometry, and *not* at equal $Re_{\max}$; the values of $U_{\max}$ and $Re_{\max}$ are tabulated for every case (Tables 1 and 2) so that the distinction is explicit at every step. This convention differs from the fixed-lid-velocity comparison of Ref. [32], in which the peak velocity was the same for all profiles and the norm differed by up to a factor of four.

2.3 Viscous dissipation and its cycle mean

For an incompressible Newtonian fluid the instantaneous viscous dissipation rate is

$$D(t) = \frac{2}{Re} \int_\Omega \mathbf{S} : \mathbf{S} \, d\Omega, \qquad \mathbf{S} = \tfrac{1}{2} \left(\nabla \mathbf{u} + \nabla \mathbf{u}^T \right), \tag{24}$$

and the quantity relevant to a periodic actuation is its cycle average

$$\overline{D} = \frac{1}{T} \int_0^T D(t) \, dt. \tag{25}$$

Splitting the velocity into a time mean and a fluctuation, $\mathbf{u} = \overline{\mathbf{u}} + \mathbf{u}'$ with $\mathbf{S} = \overline{\mathbf{S}} + \mathbf{S}'$, the cycle mean decomposes exactly as

$$\overline{D} = \underbrace{\frac{2}{Re} \int_\Omega \overline{\mathbf{S}} : \overline{\mathbf{S}} \, d\Omega}_{D(\overline{\mathbf{u}})} + \underbrace{\frac{2}{Re} \left\langle \int_\Omega \mathbf{S}' : \mathbf{S}' \, d\Omega \right\rangle}_{\text{fluctuating part}}, \tag{26}$$

because the cross term vanishes upon averaging. The two terms coincide only in the steady limit; otherwise $\overline{D} > D(\overline{\mathbf{u}})$, with a gap that measures how much of the dissipation is carried by the unsteady part of the response. Evaluating D on a time-averaged field is thus not an approximation of $\overline{D}$ with a small error: it is a different functional, and for a strongly unsteady flow it can invert the ranking of two actuations.

2.4 Mechanical energy budget

With the kinetic energy $K = \frac{1}{2} \int_\Omega |\mathbf{u}|^2 \, d\Omega$, the momentum equation gives

$$\frac{dK}{dt} = P_{\mathrm{lid}}(t) - D(t). \tag{27}$$

For a periodic state the left-hand side integrates to zero over one period, so

$$\boxed{\overline{P}_{\mathrm{lid}} = \overline{D}} \tag{28}$$

is an exact identity of the governing equations. In the numerical evaluation of Eq. (28) both terms are computed from the same discrete fields, but P_{lid} involves a wall-normal derivative at the boundary, which is evaluated with a one-sided finite difference and therefore carries a truncation error that D does not. The budget is used below as a verification of the transfer and of the periodicity of the response, with that estimator uncertainty made explicit (Sec. 6.4); it is not used as an exact closure constraint.

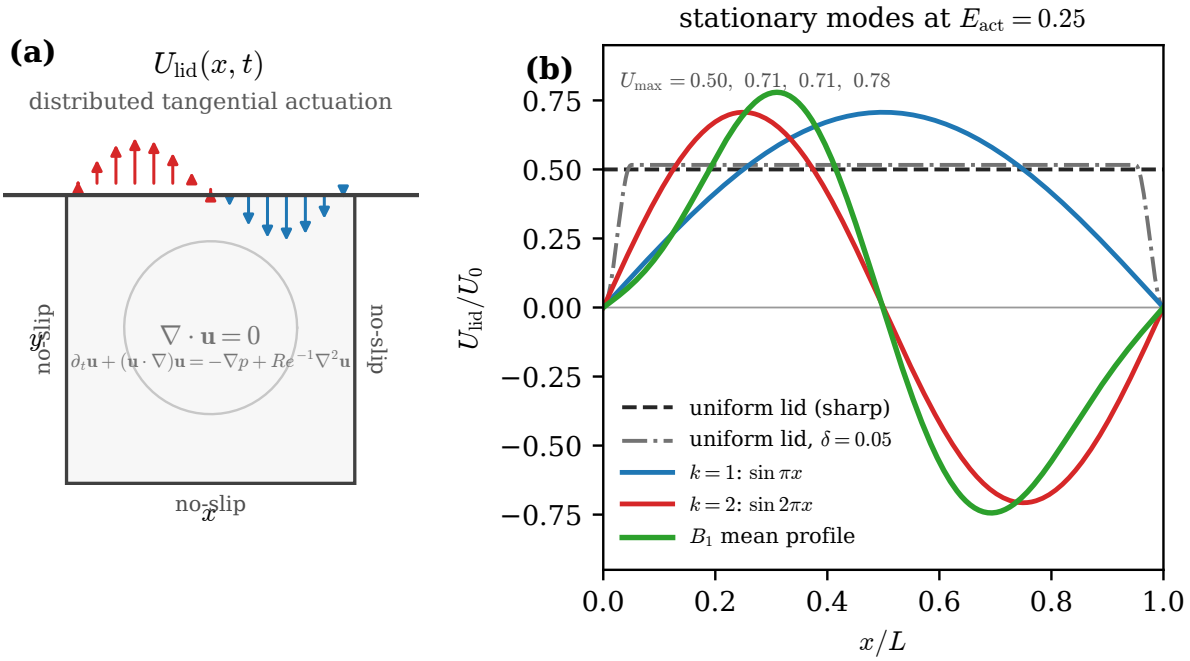


Figure 1: (a) Geometry of the controlled cavity. The upper wall moves tangentially with a distributed profile $U_{\mathrm{lid}}(x,t)$; the three other walls are no-slip and stationary. (b) The five stationary actuation profiles compared in this study, all normalized to the same quadratic actuation norm $E^*=\int_0^1 U_{\mathrm{lid}}^2\,dx=0.25$. The maximum wall velocity differs between profiles at equal norm: $U_{\max}=0.50$ for the uniform lid, 0.71 for the two harmonics and 0.78 for the B_1 mean profile.

3 Actuation modes and hydrodynamic states

3.1 Stationary reference profiles

Four stationary profiles define the comparison, all normalized to $E^*=0.25$. The *uniform* lid $U_{\mathrm{lid}}=0.5$ is the reference of the problem statement; it carries a tangential-velocity discontinuity at the two upper corners, which is retained because it is the classical boundary condition, but which is shown in Sec. 5.2 to prevent the convergence of its own dissipation. A *corner-regularized uniform* lid, with the same plateau and a raised-cosine taper of width $\delta=0.05$ at each corner

$$U_{\mathrm{lid}}=\frac{1}{2}\left[1-\cos\left(\frac{\pi x}{\delta}\right)\right]\quad(x<\delta),\tag{29}$$

and symmetrically at $x>1-\delta$, removes the discontinuity at the cost of 6.3% of the norm, which is restored by rescaling; it provides a converged reference. The first and second spatial harmonics, $U_{\mathrm{lid}}=\sqrt{2E^*}\sin(\pi x)$ and $\sqrt{2E^*}\sin(2\pi x)$, are the two elementary spatially structured controls. Finally, the time-mean profile of the identified B_1 branch (Sec. 3.2) is included as the stationary approximation of a physics-informed control.

3.2 Identified actuation branches

The time-dependent actuations are three representatives identified by the physics-informed procedure of Sec. 4 and classified by the modal energy of their control coefficients (details of the identification campaign and classification rules are provided in supplementary Tables S1–S3 and Fig. S3). Writing the lid as

$$U_{\mathrm{lid}}(x,t)=\sum_{m=1}^{N_x}\sum_{n=0}^{N_t-1}a_{mn}\sin(m\pi x)\cos(n\pi t),\tag{30}$$

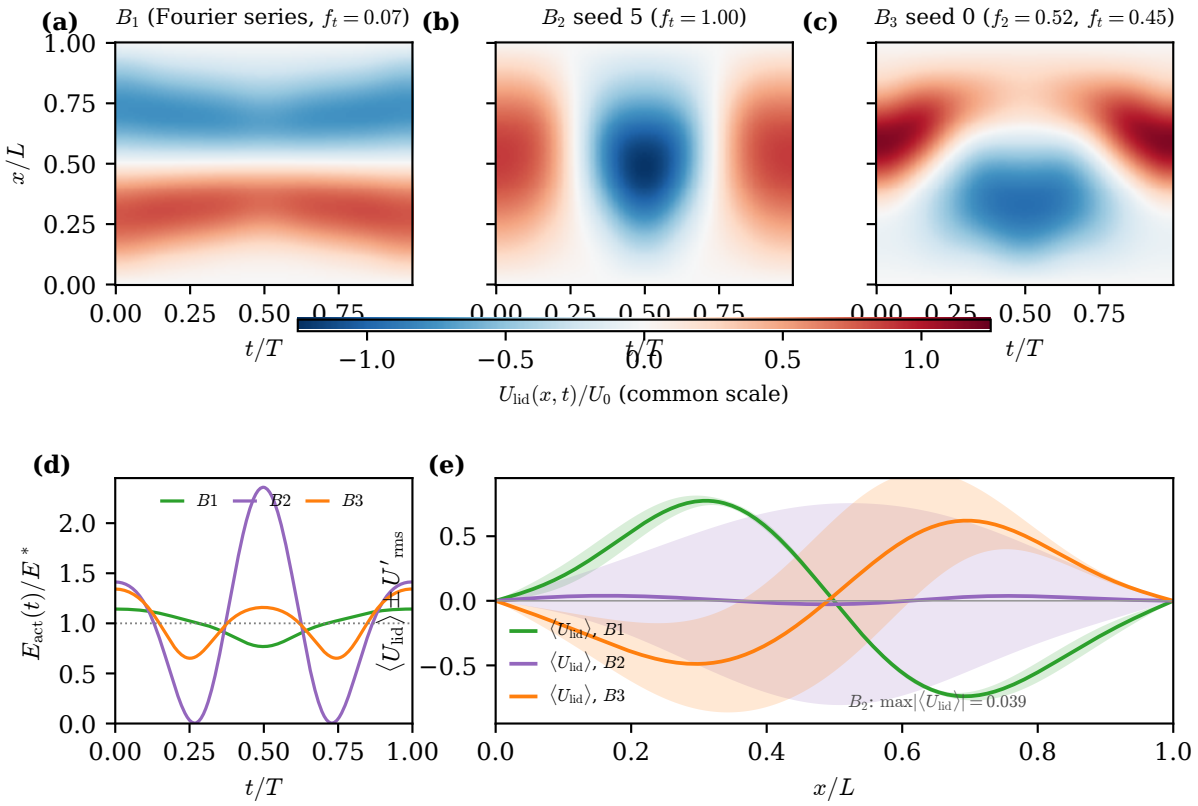


Figure 2: Spatio-temporal structure of the three identified branch controls at $Re = 500$ and $\overline{E}_{\mathrm{act}} = 0.25$. (a)–(c) Lid velocity $U_{\mathrm{lid}}(x,t)$ over one forcing period, on a common colour scale. B_1 is nearly stationary; B_2 oscillates with a full sign reversal, the wall velocity vanishing twice per period; B_3 combines a stationary two-lobe pattern with a strong oscillation. (d) Instantaneous actuation norm $E_{\mathrm{act}}(t)$ normalized by the common cycle-mean target: B_2 spans 0.004–2.36 E^*, whereas B_1 stays within 0.77–1.14 E^* and B_3 within 0.65–1.34 E^*. The comparison is therefore made at equal *cycle-mean* norm, not at equal instantaneous norm. (e) Time-mean lid profile with the temporal standard deviation as a band; the B_2 mean profile reaches only 0.039, so a mean-field diagnostic sees almost no actuation.

the energy of coefficient (m,n) is $\mathcal{E}_{m0} = \frac{1}{2}a_{m0}^2$ and $\mathcal{E}_{mn} = \frac{1}{4}a_{mn}^2$ for $n \geq 1$, with normalized fractions $f_{mn} = \mathcal{E}_{mn}/\sum \mathcal{E}_{ij}$. The stationary second-harmonic fraction is $f_2 \equiv f_{20}$ and the total time-dependent fraction is $f_t = \sum_{n\geq 1} f_{mn}$. The three classes are defined by modal dominance [32]: B_1 has $f_2 > 2f_t$ and $f_2 > 0.5$ (stationary second-harmonic dominated); B_2 has $f_t > 2f_2$ and $f_t > 0.5$ (temporally dominated); B_3 satisfies neither condition (mixed). At $Re = 500$ and $E^* = 0.25$ the representatives used here have $(f_2, f_t) = (0.93, 0.07)$ for B_1, $(0.00, 1.00)$ for B_2 (seed 5) and $(0.52, 0.45)$ for B_3 (seed 0), with a first-harmonic content $f_1 = 0.006$ for B_3 and below 0.02 for the other two.

The B_1 representative is used in two forms. The *full* series is the control as identified, including its small temporal content; the *mean profile* is its time average, a stationary actuation that isolates the spatial structure. The stationary comparison of Sec. 5 uses the mean profile and the cycle-mean comparison of Sec. 3.2 uses the full series, so that the effect of the spatial structure is never conflated with the effect of the temporal one. Representative profiles and their space–time structure are shown in Fig. 2; Table 1 collects the characteristics of the eight actuations compared in the paper.

Table 1: Actuation modes and hydrodynamic states compared in this study ($Re = 500$, fixed viscosity and geometry). $U_{\max}$ is the maximum of the time-mean lid profile and $Re_{\max} = U_{\max} Re$ the corresponding effective Reynolds number; for the two time-dependent branches the maximum over the whole period is given in parentheses. All stationary profiles and the cycle means of the time-dependent ones carry the same quadratic actuation norm, $\overline{E}_{\mathrm{act}} = 0.25$; the instantaneous norm of B_2 and B_3 varies over the period. f_2 and f_t are the stationary second-harmonic and total temporal energy fractions of the control coefficients.

Actuation	Structure	$U_{\max}$	$Re_{\max}$	f_2	f_t	Role
Uniform lid	constant, sharp corners	0.50	250	—	0	reference
Uniform lid, $\delta = 0.05$	constant, tapered	0.52	258	—	0	converged ref.
$k = 1$	$\sin \pi x$	0.71	354	1.00	0	control
$k = 2$	$\sin 2\pi x$	0.71	354	1.00	0	control
B_1 mean profile	mean of B_1	0.78	390	0.93	0	stationary
B_1 full series	Fourier series	0.78 (0.82)	388	0.93	0.07	branch
B_2 seed 5	temporal	0.04 (1.23)	20	0.00	1.00	unsteady
B_3 seed 0	mixed	0.62 (1.16)	311	0.52	0.45	intermediate

4 Numerical methodology

4.1 Identification of the actuation

The control coefficients $a_{mn}(Re)$ and the flow field are obtained simultaneously by a parametric physics-informed neural network with inputs $(x, y, t, \widetilde{Re})$ and a multi-objective loss combining the momentum and continuity residuals, the boundary conditions, a viscous-dissipation objective, the actuation-norm constraint $(E_{\mathrm{act}} - E^*)^2$, a spatial-variance regularizer and a Reynolds-separation term. The identification is performed once over $Re \in [100, 1000]$; the branch representatives at $Re = 500$, $E^* = 0.25$ are then exported on one forcing period and transferred without retraining. The network architecture, the sampling protocol, the loss weights, the training schedule and the branch-classification rules are reported in the Supplementary Material (Secs. S1–S2) and in Ref. [32]; the present paper uses the identified controls as physical boundary data and does not re-examine the identification procedure. What matters here is only that the exported profile is the one the optimizer produced, without harmonic substitution or spatial idealization, and that its realized norm is the prescribed one to within 0.16%.

4.2 Independent MRT-LBM evaluation

The flows are computed with a multiple-relaxation-time D2Q9 lattice-Boltzmann solver [26, 27] that is independent of the identification framework: a Cartesian lattice instead of scattered collocation, a different discretization of the same equations, and a different iteration strategy. Collision is performed in moment space with a diagonal relaxation matrix, and the moving lid is imposed with the Zou–He velocity boundary condition [28, 29]. The solver reproduces the benchmark centerline velocities of Ghia et al. [1] with $L_2(u) = 0.0051$ and $L_\infty(u) = 0.0128$ at $Re = 100$, and $L_2(u) = 0.0141$ and $L_\infty(u) = 0.0300$ at $Re = 1000$ on the 256×256 lattice (Supplementary Table S1), which fixes the resolution used for the controlled flows.

Each actuation is advanced from rest with the lid switched on at $t = 0$. At least 200 forced periods are discarded, a drift test on the cycle-averaged dissipation fixes the end of the settling phase, and 64 consecutive periods are retained for the statistics. The same protocol is used for every case, so the comparison does not depend on the transient length. Instantaneous dissipation, kinetic energy and lid power are accumulated at every step; the cycle mean is a direct accumulation over the 64 sampled periods, not an eight-point phase quadrature, because the latter is 5% low for the most unsteady responses. The wall shear in Eq. (23) is evaluated with a first-order one-sided derivative of the recovered wall velocity, and the resulting estimator uncertainty is quantified in Sec. 6.4.

Table 2: Stationary controls at $Re = 500$ and matched quadratic actuation norm $E^* = 0.25$ on three grids. ε_N is the integrated viscous dissipation; p_{obs} and GCI_{512} are the observed order and the fine-grid grid-convergence index of the absolute dissipation ($F_s = 1.25$). The last column gives the dissipation normalized by the sharp and by the corner-regularized uniform lid on the same grid (the two values are separated by a slash). The B_1 mean profile is used as computed ($E_r = 0.2503$). A dagger marks a flagged triple, for which the extrapolated value is not used. $U_{\max}$ and $Re_{\max}$ are repeated here because the norm constraint fixes the rms wall velocity (0.5) but not the peak.

Control	$U_{\max}$	$Re_{\max}$	ε_{128}	ε_{256}	ε_{512}	p_{obs}	GCI_{512} (%)	$\varepsilon_{512}/\varepsilon_{512}^{\mathrm{ref}}$ (sharp / reg.)
Uniform lid (sharp)	0.50	250	0.00774	0.00948	0.01102	0.18	$130.71^{\dagger}$	1.000 / 1.272
Uniform lid, $\delta = 0.05$	0.52	258	0.00717	0.00815	0.00866	0.96	7.79	0.786 / 1.000
$k = 1$ ($\sin \pi x$)	0.71	354	0.00635	0.00688	0.00713	1.09	3.85	0.647 / 0.823
$k = 2$ ($\sin 2\pi x$)	0.71	354	0.01012	0.01129	0.01185	1.04	5.62	1.076 / 1.368
B_1 mean profile	0.78	390	0.01067	0.01189	0.01250	1.01	6.01	1.134 / 1.443

4.3 Grid convergence and uncertainty

The three stationary controls and both uniform references are computed on $N = 128$, 256 and 512 lattices. The observed order of convergence p_{obs} and the fine-grid grid-convergence index are obtained by Richardson extrapolation with a safety factor $F_s = 1.25$ [30, 31]. A triple is flagged, and its extrapolated value discarded, when it is non-monotone, when $p_{\mathrm{obs}} \notin [0.1, 4]$, or when the GCI exceeds 100%: in those cases the smooth-error assumption underlying the estimator is not credible. The estimator is applied both to the absolute dissipation and to the dissipation ratios, since the ratios are the quantities used in the physical discussion and their errors need not cancel. The time-dependent branches are ranked on a single $N = 256$ lattice, with two realizations per branch; that restriction is stated wherever those numbers are used.

5 Energetic response of stationary wall-actuation modes

5.1 Comparison at matched actuation norm

Table 2 and Fig. 3 give the integrated dissipation of the five stationary controls at $Re = 500$ and $E^* = 0.25$. On the finest grid the first harmonic is the only control that lowers the dissipation relative to the uniform lid of the problem statement: $\varepsilon_{512} = 0.00713$ against 0.01102, a ratio of 0.647. The second harmonic raises it to 0.01185 (ratio 1.076) and the B_1 mean profile to 0.01250 (ratio 1.134). The ordering $k = 1 <$ uniform $< k = 2 \approx B_1$ is the central stationary result, and it is the reverse of the fixed-lid-velocity ordering reported in Ref. [32], where both harmonic profiles were less dissipative than the uniform lid and the second harmonic was the least dissipative of the three. The reversal is not a refinement of that result but a consequence of the constraint: at equal lid speed the harmonics have half the norm of the uniform lid, and restoring the norm raises their wall velocity by $\sqrt{2}$ and their effective Reynolds number with it.

The ratios built on the sharp uniform lid are not converged estimates of the shape effect, because the reference itself does not converge: its increments between successive grids decay by only 12% per refinement ($p_{\mathrm{obs}} = 0.18$, GCI 131%), and the formal Richardson limit, 2.25×10^{-2}, is more than twice the finest-grid value and is not a credible continuous-limit dissipation. The three other controls converge, with $p_{\mathrm{obs}} = 1.09$, 1.04 and 1.01 and fine-grid GCIs of 3.9%, 5.6% and 6.0%. Normalizing by the corner-regularized lid instead gives ratios of 0.823, 1.368 and 1.443 at $N = 512$, i.e. -17.7%, $+36.8\%$ and $+44.3\%$, with Richardson estimates 0.802, 1.347 and 1.427 and ratio GCIs of 3.1%, 1.9% and 1.4%. The sign of the ordering is the same under both references and at all three resolutions; only its magnitude changes, because the non-converged corner contribution inflates the denominator of the sharp-lid ratios. We therefore quote the regularized ratios as the better-defined estimate of the shape effect and the sharp-lid ratios as the values directly comparable with the classical uniform-lid baseline, and we do not use the sharp-lid ratio for $k = 1$

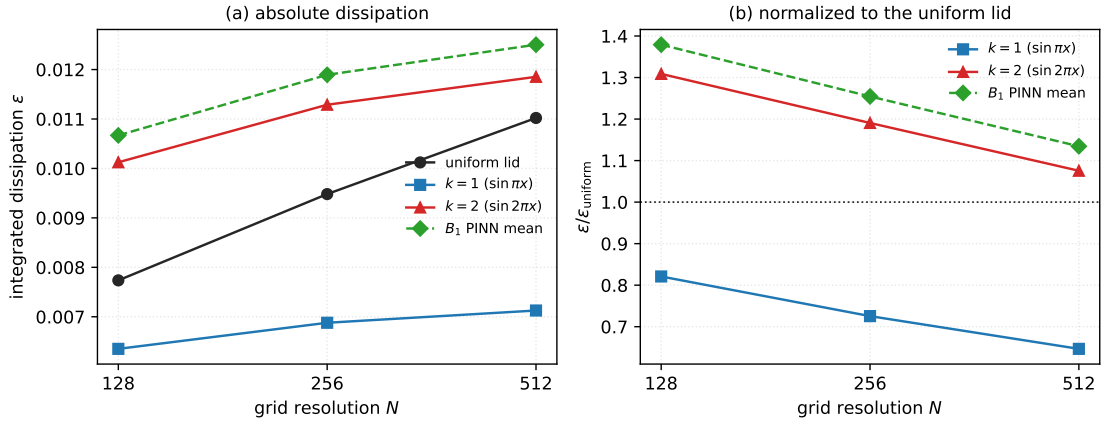


Figure 3: Grid convergence of the stationary comparison at $Re = 500$ and matched actuation norm $E^* = 0.25$. (a) Integrated dissipation versus resolution for the four stationary controls and for the corner-regularized uniform lid. (b) Dissipation normalized by the sharp uniform lid on the same grid: this reference does not converge, and the ratios drift by 18–21% over the triple. (c) Dissipation normalized by the regularized lid: the three ratio triples are monotone and converge with GCIs of 1.4–3.1%. The sign of the ordering $k = 1 <$ uniform $< k = 2 \approx B_1$ is observed at every resolution in both (b) and (c); its magnitude is reference-dependent.

as a quantitative reduction: -35.3% overstates the -17.7% obtained against a converged reference. The separation between $k = 2$ and the B_1 mean profile (1.368 against 1.443, about 5%) exceeds their GCIs but not by a wide margin, so they are reported as approximately equal rather than strictly ordered.

5.2 The corner singularity and the reference dependence

The non-convergence of the uniform lid is a property of its corners, not of the uniform profile as such. Its dissipation grows by 16% between $N = 256$ and $N = 512$, the signature of a boundary-layer singularity that the grid cannot resolve. Replacing the discontinuity by the cosine taper of Eq. (29) removes 21% of the dissipation at $N = 512$ and, more importantly, changes the growth between the two finest grids from 16% to 6%, i.e. the expected first-order rate: $p_{\mathrm{obs}} = 0.96$, GCI 7.8%, extrapolated $\varepsilon_\infty = 9.20 \times 10^{-3}$. A continuous lid therefore converges, and the singular one does not. This has two consequences for the interpretation. First, a comparison of actuation shapes that uses the discontinuous lid as denominator mixes the shape effect with a resolution artefact; the regularized reference is the only one that separates them. Second, the corner is an energetic feature in its own right: the uniform lid concentrates a large part of its dissipation in two small regions whose contribution grows with resolution, so any ranking that depends on the corner peak is resolution-dependent by construction.

The practical conclusion is that two numbers must be reported for each profile, not one. The sharp-lid ratio answers the question "what does this actuation do relative to the classical cavity?" and is the quantity a reader will compare with the literature. The regularized ratio answers "what is the effect of the spatial shape at fixed actuation norm?" and is the quantity that supports a physical statement about shape. Both are given in Table 2, and the text uses the second for every claim about ordering or magnitude.

5.3 Local mechanism of the second-harmonic penalty

The ranking has a local origin that can be read directly from the fields (Fig. 4). The uniform lid concentrates its dissipation in the two upper corners, where the imposed tangential velocity jumps to zero: the peak strain-rate magnitude there is 169 in units of U_0/L, against 15.0 for $k = 1$ and 23.4 for $k = 2$, and those corner layers are the brightest features of the dissipation field. Both

harmonics remove the jump, and their dissipation is spread over the lid shear layer and the side walls instead. But the second harmonic pays for this in the interior. Its lid generates a pair of counter-rotating cells of nearly equal strength, separated by a wall-to-wall separatrix at $x = 0.5$, each enclosing its own low-strain core; the uniform and first-harmonic lids recover the classical single primary vortex with a corner eddy. The reorganization shifts the strain-rate distribution towards larger values: the interior mean $\langle|\mathbf{S}|\rangle$ is 1.450 for $k = 2$, against 1.039 for the uniform lid and 1.088 for $k = 1$, a 40% increase at identical actuation norm.

The trade-off is explicit. The first harmonic removes almost all of the corner peak (169 to 15.0) at a cost of 4.7% in interior mean strain, and is a net reduction. The second harmonic removes slightly less of it (23.4) while raising the interior mean by 39.6%, and is a net penalty: it buys a symmetric two-cell state with a stronger interior strain field, and the corner singularity it removes is not costly enough to compensate. The second harmonic therefore pays roughly 40% more interior strain to avoid a corner peak that contributes less than 5% of the total dissipation. Additional parametric studies (network capacity, loss-term ablation, aspect-ratio dependence, and Fourier vs. Chebyshev bases) are reported in supplementary Secs. S3–S7 and Figs. S5–S7; they confirm that the ordering is robust to the numerical parameters and is a physical property of the flow response to the actuation structure. eight times the interior penalty of the first for no additional corner gain. This is the mechanical content of the +36.8% and +44.3% of Table 2: the energetic cost of a spatial mode is set by the volume over which it raises the strain, not by the peak it removes.

6 Cycle-mean dissipation of time-dependent actuation branches

6.1 Why the dissipation of the mean flow is not the cycle mean

For the stationary controls of Sec. 5 the distinction between $D(\overline{\mathbf{u}})$ and $\overline{D}$ is empty, because the response is steady. For the time-dependent branches it becomes the central issue. Equation (26) is exact: the cycle mean exceeds the mean-field value by the dissipation of the fluctuations, and there is no reason for that term to be small when the actuation reverses sign. A mean-field diagnostic answers the question "how much would this flow dissipate if it were steady?"; it does not answer "how much does it dissipate?". Figure 5 and Table 3 show how far apart the two answers can be; supplementary Table S5 and Fig. S8 provide additional temporal statistics and phase-resolved analyses for the three branches.

6.2 Cycle-mean comparison of the three branches

Table 3 collects the result. The five quasi-stationary controls span less than a factor of two, from 6.88×10^{-3} for $k = 1$ to 1.22×10^{-2} for the full B_1 series, and their mean-field and cycle-mean values coincide to within 4%; for the B_1 series the difference is 1.17×10^{-2} against 1.22×10^{-2}, which is the residual effect of its small temporal content. The branches with substantial temporal content leave that band. B_2 dissipates 3.76×10^{-2} over the cycle while its mean field dissipates 3.10×10^{-5}, a ratio of 1214; its mean field holds only 6% of the cycle-mean kinetic energy, so the mean motion is a shallow recirculation under the lid, not a weakened copy of the instantaneous flow. B_3 is intermediate: $D(\overline{\mathbf{u}}) = 5.05 \times 10^{-3}$ and $\overline{D} = 2.18 \times 10^{-2}$, a ratio of 4.3. On the cycle average, and at the same actuation norm, B_2 costs 3.1 times B_1 and 5.5 times the least dissipative stationary control; its two realizations agree to 1.7%, whereas the two B_3 realizations differ by 23%, so the B_3 value quoted here is the lower of the two and must be read as a lower bound on that class.

Three statements follow, and they are the ones the paper retains. First, the mean-field functional is not a surrogate for the cycle mean: it can be three orders of magnitude too small and it reverses the ranking of the branches. Second, the temporal fraction f_t orders the branches by accessibility, not by cost: B_2, the most strongly time-dependent control, is the most dissipative of

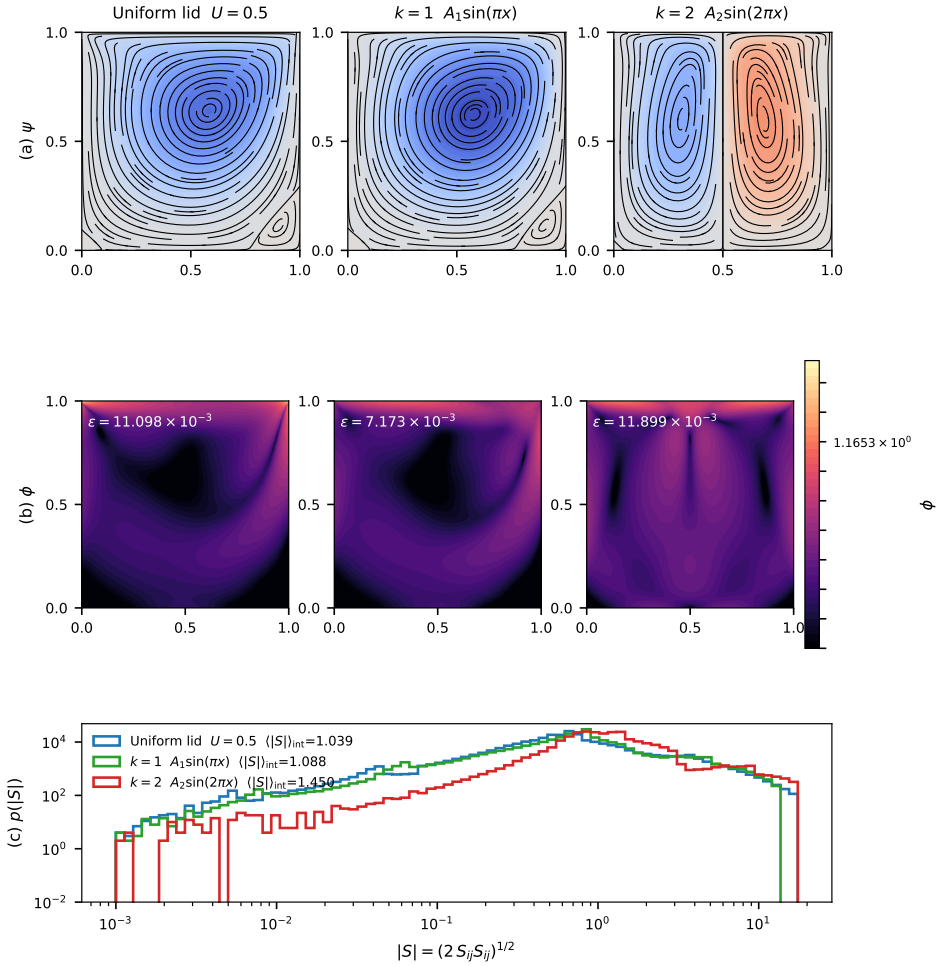


Figure 4: Local origin of the stationary ranking at $Re = 500$, $N = 512$ and matched actuation norm, on scales common to the three columns. (a) Stream function of the steady response with the streamline topology. (b) Local viscous dissipation $\phi = \nu(2S_{11}^2 + 2S_{22}^2 + 4S_{12}^2)$ on a logarithmic scale spanning six decades below the global maximum; the annotation gives the domain integral. (c) Probability density of the strain-rate magnitude over the cavity interior (three boundary cells excluded), with the interior mean $\langle|\mathbf{S}|\rangle_{\rm int}$. The uniform lid peaks at the corners ($|\mathbf{S}|_{\max} = 169$); $k = 1$ removes the peak at almost no interior cost ($\langle|\mathbf{S}|\rangle_{\rm int} = 1.088$ against 1.039); $k = 2$ removes less of it (23.4) and raises the interior mean to 1.450, which is the origin of its dissipation penalty.

the three at matched norm. Third, the ranking of the time-dependent branches ($B_1 < B_3 < B_2$) is the same as the ranking of their mean-field dissipation read in reverse, which is a direct consequence of Eq. (26): the branch that suppresses its mean flow most effectively is the one that pays most in fluctuations.

6.3 Flow organization behind the cycle-mean cost

The field-level organization explains the numbers (Figs. 6 and 7). The B_1 time-mean flow is the familiar pair of counter-rotating cells centred at $(0.31, 0.54)$ and $(0.69, 0.56)$, with $\max|\langle\mathbf{u}\rangle| = 0.733$ and $\langle\psi\rangle \in [-0.036, 0.035]$; half of its mean kinetic energy lies below $y = 0.675$, so the cells fill the cavity. Its instantaneous states coincide with that mean to 2.7% in field norm and its kinetic-energy fluctuation ratio is $R_K = 4 \times 10^{-4}$, so it is quasi-stationary in the operational sense. The B_2 mean field is 19 times weaker in stream function ($\max|\langle\psi\rangle| = 0.0019$) and 18 times weaker in peak speed (0.042, or 5.7% of the B_1 value); its mean extrema sit at $y = 0.89$–0.92 and half of the mean kinetic

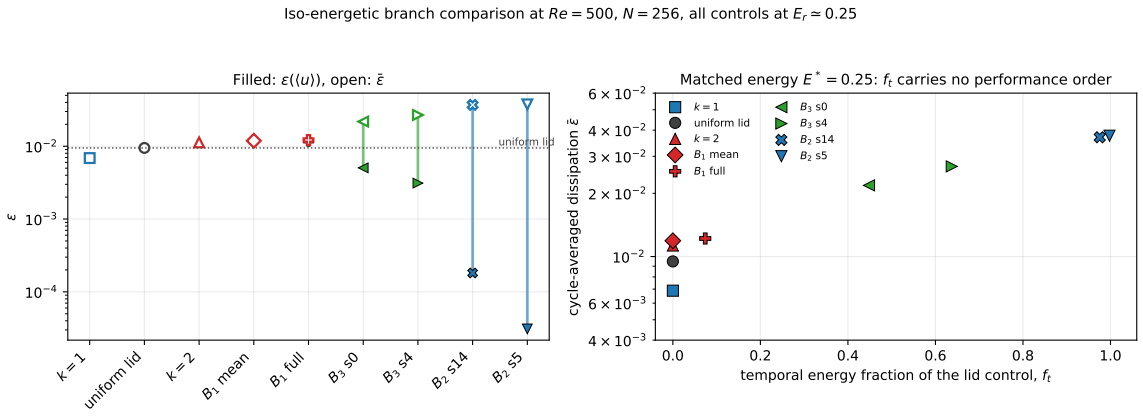


Figure 5: Iso-energetic comparison of the nine controls at $Re = 500$, $\overline{E}_{\mathrm{act}} = 0.25$ and $N = 256$. Left: the two dissipation functionals of each control, the dissipation of the time-mean field (filled markers) and the cycle mean (open markers), joined by the gap between them; the two coincide for stationary controls. Right: cycle-mean dissipation against the temporal energy fraction f_t of the lid control. The ordering in f_t is the order in which the branches are recovered by the identification, and it is not the order of the dissipation cost.

Table 3: Dissipation of the time-mean field and cycle-mean dissipation at $Re = 500$, $N = 256$ and cycle-mean actuation norm $\overline{E}_{\mathrm{act}} = 0.25$. $D(\overline{\mathbf{u}})$ is the functional evaluated on the time-mean field; $\overline{D}$ is the average of the instantaneous dissipation over the phase-resolved periodic response and is the performance index relevant to a periodic flow. The two coincide only in the steady limit. A_ε is the half-spread of $D(t)$ over the cycle, i.e. the variability concealed by a single cycle-mean number. The B_1 mean profile row is stationary by construction; the B_1 full row is the identified series.

Control	f_t	$\overline{E}_{\mathrm{act}}$	$D(\overline{\mathbf{u}})$	$\overline{D}$	$\overline{D}/D(\overline{\mathbf{u}})$	A_ε
Uniform lid	0.000	0.250	9.48×10^{-3}	9.48×10^{-3}	1.00	0
$k = 1$ ($\sin \pi x$)	0.000	0.250	6.88×10^{-3}	6.88×10^{-3}	1.00	0
$k = 2$ ($\sin 2\pi x$)	0.000	0.250	1.13×10^{-2}	1.13×10^{-2}	1.00	0
B_1 mean profile	0.000	0.250	1.19×10^{-2}	1.19×10^{-2}	1.00	0
B_1 full series	0.074	0.250	1.17×10^{-2}	1.22×10^{-2}	1.04	2.4×10^{-3}
B_2 seed 5	0.998	0.250	3.10×10^{-5}	3.76×10^{-2}	1214	3.6×10^{-2}
B_2 seed 14	0.975	0.250	1.82×10^{-4}	3.70×10^{-2}	204	3.5×10^{-2}
B_3 seed 0	0.448	0.250	5.05×10^{-3}	2.18×10^{-2}	4.3	1.5×10^{-2}
B_3 seed 4	0.637	0.250	3.12×10^{-3}	2.69×10^{-2}	8.6	2.6×10^{-2}

energy lies above $y = 0.95$. Its streamwise fluctuation, by contrast, reaches 0.555—thirteen times its own mean and 76% of the B_1 mean peak. The B_3 flow keeps an intermediate mean circulation ($\max|\langle \mathbf{u} \rangle| = 0.595$, 81% of B_1) but places it close to the lid, with extrema at $(0.20, 0.86)$ and $(0.77, 0.81)$ and half of the mean kinetic energy above $y = 0.91$; its fluctuation reaches 0.416.

"Weak mean flow" is therefore not "weak flow". The controls with a strong temporal component do not produce a small perturbation around a mean recirculation: they produce responses whose fluctuations dominate the instantaneous field while the mean is nearly zero. A figure that normalized each panel by its own maximum, or a diagnostic that used only the mean field, would hide exactly the contrast that defines the two unsteady branches.

6.4 Mechanical energy budget of the branches

Table 4 and Fig. 8 close the comparison with the budget of Eq. (27). The kinetic energy separates the branches by two orders of magnitude (1.15×10^{-2}, 2.47×10^{-4} and 3.98×10^{-3}), while the cycle-mean dissipation orders them the other way round (1.22×10^{-2}, 3.76×10^{-2} and 2.18×10^{-2}). Within the period, $D(t)$ varies by a factor 1.5, 10.9 and 4.2 respectively, and the lid power of B_2

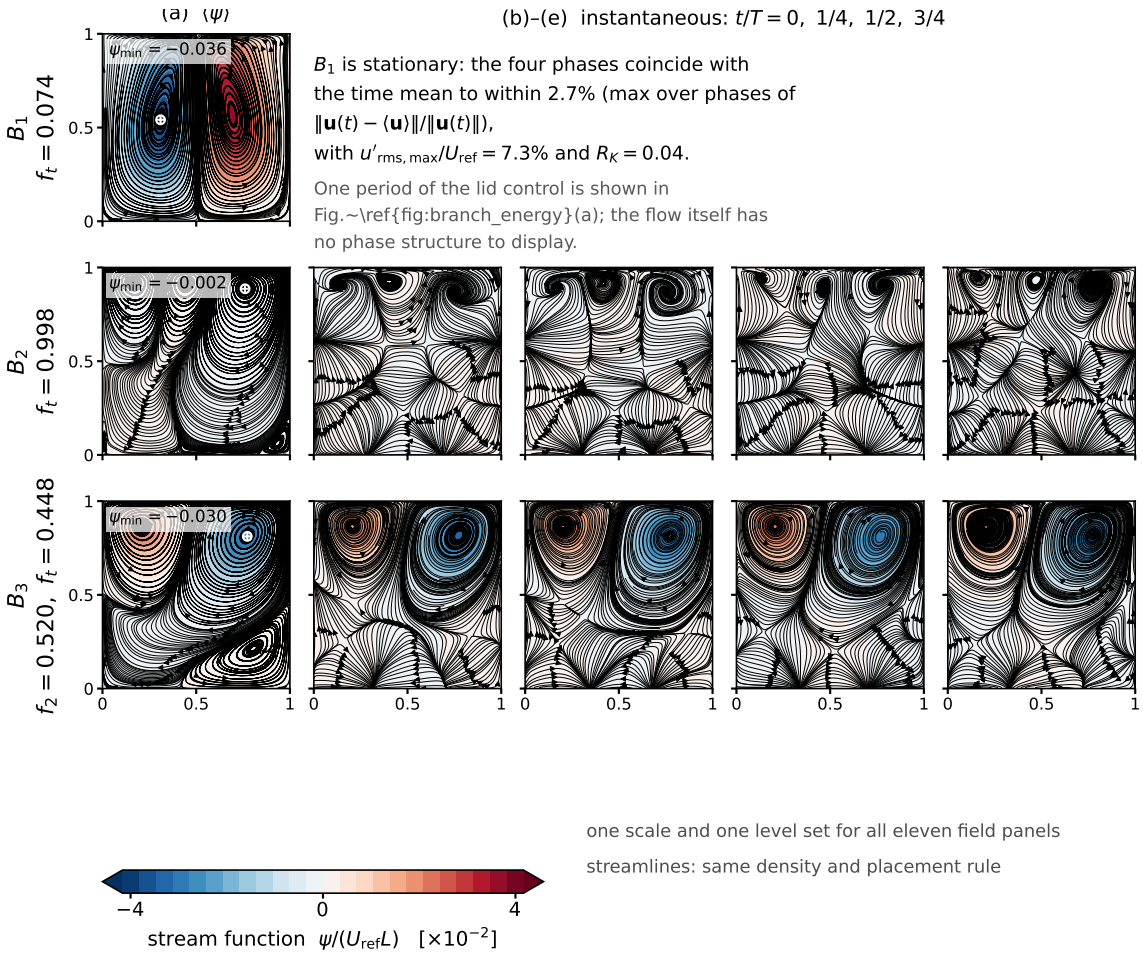


Figure 6: Flow topology of the three branch controls at $Re = 500$, $E^* = 0.25$ and $N = 256$, on one stream-function scale and one set of contour levels common to all panels. Rows: B_1 (full series, $f_t = 0.074$), B_2 (seed 5, $f_t = 0.998$) and B_3 (seed 0, $f_2 = 0.520$, $f_t = 0.448$). (a) Time mean $\langle\psi\rangle$ with the streamlines of the mean flow; the circle marks its negative extremum. (b)–(e) Instantaneous ψ and streamlines at $t/T = 0, 1/4, 1/2, 3/4$ for B_2 and B_3; the four phases belong to the same periodic state, whose first- and second-half phase averages differ by at most 1.0×10^{-3} in relative field norm. For B_1 the instantaneous states coincide with the mean to 2.7%, so the four panels are replaced by that statement. The common scale is deliberate: the B_2 mean cell is 19 times weaker than the B_1 one and is nearly invisible, which is the point.

is negative for 14% of the period: the flow returns energy through the lid while the Stokes layer spins down. The residual $\overline{P}_{\mathrm{lid}} - \overline{D}$ is -2.6%, $+5.6\%$ and $+3.8\%$.

The residual is two to three orders of magnitude larger than the drift of K over the sampling window, which is at most $1.5 \times 10^{-2}\%$ of $\overline{D}$; it is therefore not a failure of periodicity. It is the truncation error of the one-sided wall-shear estimate. Two checks bound that error. First, the first-order estimator reproduces the independently accumulated P_{lid} of the same fields to 0.3% for B_1, so it is internally consistent; the residual is a property of the estimator, not of the accumulation. Second, replacing it by a second-order one-sided derivative on the same fields moves the B_1 mean-field residual from -2.87% to $+0.58\%$, i.e. it crosses zero, which shows that the estimator uncertainty is of the order of the residual itself (a few per cent). Applied to the branch cycle means, the same change moves the B_1 residual to $+1.4\%$ but inflates the B_2 and B_3 residuals, because the three-point stencil amplifies lattice-scale noise in the thin Stokes layer. We therefore retain the first-order estimator for all three branches, report the residual as measured, and present Eq. (28) as an energetic verification with a few-per-cent estimator uncertainty rather than as an exact closure. This is sufficient for the conclusions drawn here, which rest on differences

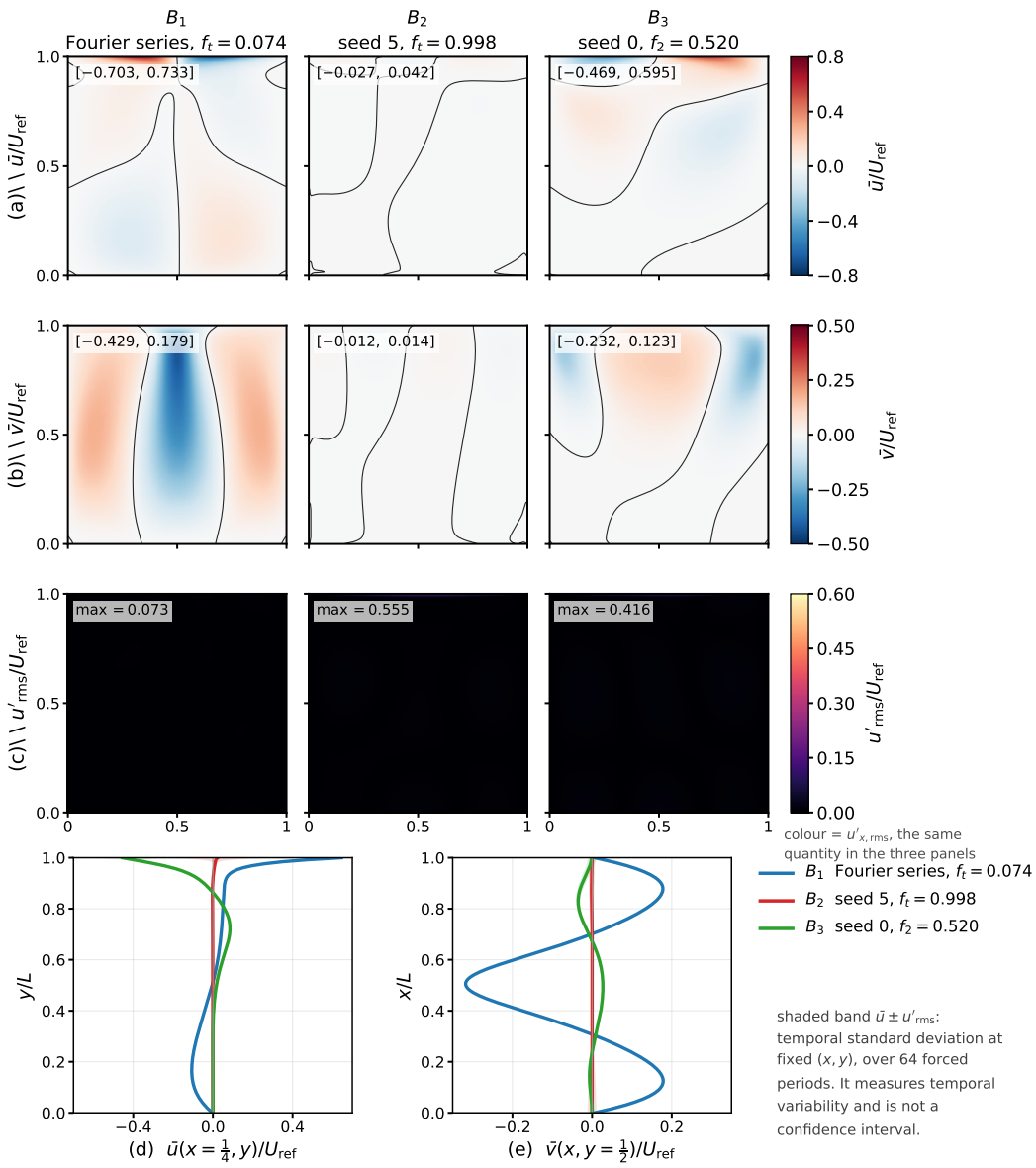


Figure 7: Velocity fields and centreline profiles of the same three captures. Rows: (a) $\bar{u}$, (b) $\bar{v}$ and (c) $u'_{x,\mathrm{rms}}$, each on one colour range for the three columns and a single normalization, with the occupied range written in every panel so that the common scale can be checked. Grey lines are the zero contours of the mean fields. (d) $\bar{u}$ along $x = 1/4$ with the $\pm u'_{\mathrm{rms}}$ band; the midline $x = 1/2$ is not used because the antisymmetric controls have $u \equiv 0$ there. (e) $\bar{v}$ along $y = 1/2$, with $\bar{v}$ on the abscissa. The bands are temporal standard deviations over the sixty-four sampled periods, not confidence intervals.

of a factor of three, not on the third significant digit.

7 Physical interpretation and implications for flow control

7.1 Why the stationary second harmonic increases dissipation

At matched quadratic actuation norm the second harmonic is not a low-dissipation mode, and the reason is geometric rather than spectral. The lid profile $A_2 \sin(2\pi x)$ imposes two opposite lobes of wall velocity, which drive two counter-rotating cells of comparable strength instead of one primary vortex. That reorganization has two competing energetic consequences. It removes the corner velocity jump, which suppresses a singular strain peak that dominates the dissipation

Table 4: Mechanical energy budget of the three branch controls at $Re = 500$ and $N = 256$, at the common cycle-mean actuation norm $E^* = 0.25$: B_1 is the full series ($f_t = 0.074$), B_2 is seed 5 ($f_t = 0.998$) and B_3 is seed 0 ($f_2 = 0.520$, $f_t = 0.448$). All quantities are cycle averages over 64 forced periods of the instantaneous fields; $\overline{D}$ is never replaced by $D(\overline{\mathbf{u}})$. $\sigma(\overline{D})$ is the block standard error over eight blocks of eight cycles, expanded by $t_{7,0.975} = 2.365$; it is below 0.1% for all three branches because the response is periodic to better than 10^{-3}, so the physically relevant uncertainty is the seed-to-seed spread (1.7% within B_2, 23% within B_3). The residual $\overline{P}_{\mathrm{lid}} - \overline{D}$ is the first-order truncation of the wall-shear estimator, not a physical imbalance (Sec. 6.4).

Branch	$\overline{E}_{\mathrm{act}}$	$\overline{K}$	$\overline{P}_{\mathrm{lid}}$	$\overline{D}$	$\sigma(\overline{D})$
B_1 (full series)	0.2500	1.15×10^{-2}	1.19×10^{-2}	1.22×10^{-2}	$< 0.1\%$
B_2 (seed 5)	0.2504	2.47×10^{-4}	3.97×10^{-2}	3.76×10^{-2}	$< 0.1\%$
B_3 (seed 0)	0.2503	3.98×10^{-3}	2.26×10^{-2}	2.18×10^{-2}	$< 0.1\%$

of the uniform lid. It also forces the return flow through a wall-to-wall separatrix at the cavity centre, which raises the strain over the whole interior: the interior mean of $|\mathbf{S}|$ increases by 40% at identical norm. The second effect wins, because the corner layer is thin while the interior is not. The first harmonic removes the same corner peak at almost no interior cost, and is the only stationary profile that is a net reduction. The energetic cost of a spatial actuation mode is thus controlled by the volume over which it raises the strain, not by the peak it removes; a mode that is "smoother" at the wall need not be less dissipative in the bulk.

7.2 Why temporal forcing increases the cycle-mean dissipation

For a periodic control the relevant functional is the cycle mean of a quadratic quantity, and the fluctuations are a first-order contribution to it, not a correction. Equation (26) makes this explicit, and the B_2 branch is its clearest illustration: a mean field that dissipates 3.1×10^{-5} and a cycle mean that dissipates 3.8×10^{-2}, with the mean field holding 6% of the kinetic energy. The physical reason is that the oscillating lid generates a Stokes layer whose wall-normal gradients are large and reverse twice per period; the instantaneous strain is therefore large wherever the instantaneous velocity is large, and the time average of its square cannot be recovered from the average velocity. The same mechanism explains why the lid power of B_2 is negative for 14% of the period: while the Stokes layer spins down, the flow pushes the wall and returns part of the energy that it received earlier. A control designer who evaluates such a state from its mean field will conclude that it is almost frictionless, whereas the cycle average ranks it as the most dissipative of the three branches.

7.3 What branch accessibility does and does not imply

The three branches were not selected by their dissipation. They were selected by a physics-informed optimization whose loss contains a dissipation objective, an actuation-norm constraint and a spatial-variance regularizer, and whose outcome depends on the initialization and on the regularization weight [32]. At $Re = 500$ and $E^* = 0.25$ the stationary second-harmonic structure is the most frequently recovered one, and the full B_1 series is quasi-stationary and reproducible; that is a statement about the variational landscape, and it is confirmed here by the independent solver. It is not a statement about efficiency. At the same actuation norm the branch most frequently recovered is a net dissipation penalty relative to the first harmonic, and the branch with the strongest temporal content, which is also recovered in a reproducible way, is the most dissipative of the three. The operational conclusion is that accessibility and physical performance are different properties, and that a control study which reports only the recovered mode, or only its mean-field dissipation, can invert the energetic ranking of the actuations it compares.

Two limits of this statement must be stated explicitly, because they define its scope. First, the temporal comparison covers three representative realizations at a single Reynolds number

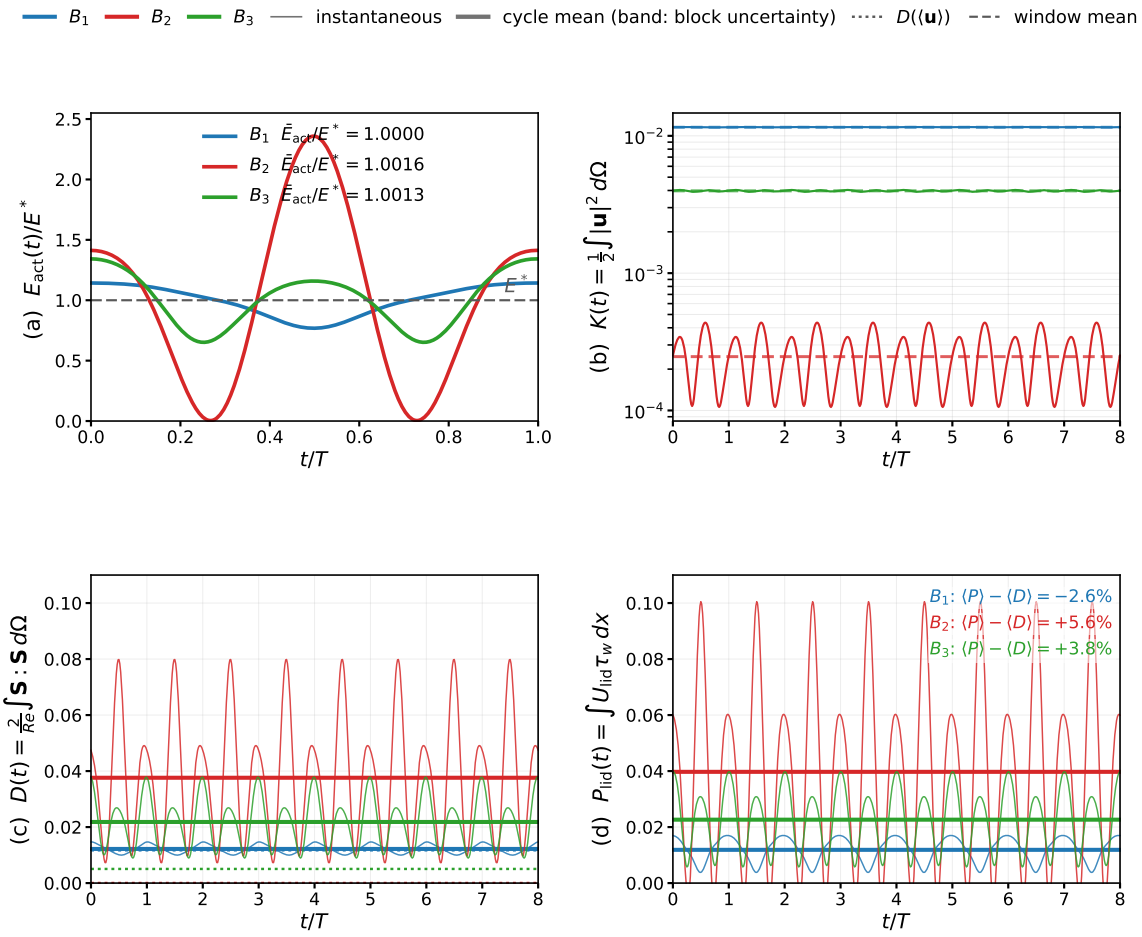


Figure 8: Phase-resolved energetic response of the three branches at $Re = 500$ and $E^* = 0.25$. (a) Instantaneous actuation norm $E_{\mathrm{act}}(t)$ over one forced period, normalized by the common cycle-mean target. (b) Kinetic energy $K(t)$ over the first eight of the sixty-four sampled periods (logarithmic ordinate). (c) Viscous dissipation $D(t)$: instantaneous (thin), cycle mean (thick, with the block uncertainty as a band) and, dotted, the dissipation $D(\langle\mathbf{u}\rangle)$ of the time-mean field, which is not substituted for the cycle mean. (d) Lid power $P_{\mathrm{lid}}(t)$, positive into the fluid, with the same conventions; the annotations give the window-averaged residual $\overline{P}_{\mathrm{lid}} - \overline{D}$. The kinetic-energy balance integrates to zero over one period of a periodic state, so the two cycle means must agree; the measured residual is the wall-shear estimator error discussed in the text.

($Re = 500$) and a single grid ($N = 256$), with two realizations for B_2 and B_3; the within-class spread is 1.7% for B_2 but 23% for B_3, so the B_3 cost is a lower bound on its class. The stationary comparison, by contrast, rests on three grids and two reference lids. Second, the B_1 temporal content is small but not zero ($f_t = 0.074$), and its cycle mean exceeds its mean-field value by 4%; the quasi-stationary limit is approached, not attained. Generalization to other Reynolds numbers, to three-dimensional cavities, or to actuations with a different spatial support is not implied by the present results.

8 Conclusions

Four statements summarize the study.

1. *The spatial structure of wall actuation reorganizes the cavity and the strain field.* At matched quadratic actuation norm the second spatial harmonic replaces the single primary vortex of the uniform lid by two counter-rotating cells and raises the interior mean strain rate by 40%; the first harmonic removes the corner strain peak at almost no interior cost. The corner

singularity of the discontinuous lid is an energetic feature in its own right: it prevents the convergence of the reference dissipation ($p_{\mathrm{obs}} = 0.18$, GCI 131%) and disappears once the corner is regularized ($p_{\mathrm{obs}} = 0.96$, GCI 7.8%).

2. *The ranking of stationary profiles depends on the constraint.* At equal actuation norm, and relative to a converged corner-regularized reference, the first harmonic reduces the dissipation by 17.7%, while the second harmonic and the mean profile of the identified stationary branch increase it by 36.8% and 44.3%. At equal lid speed the same three profiles are all less dissipative than the uniform lid and the second harmonic is the best of them [32]; the inversion is produced by the norm constraint, which fixes the rms wall velocity but not the peak, and therefore not the effective Reynolds number ($Re_{\max} = 250$ for the uniform lid, 354 for the harmonics, 390 for the B_1 mean profile at $E^* = 0.25$).

3. *The cycle-mean dissipation of a time-dependent actuation is not the dissipation of its mean field.* For the temporally dominated branch the two differ by a factor 1214, and the branch that is most strongly damped in the mean is the most dissipative on the cycle (3.76×10^{-2}, i.e. 3.1 times the quasi-stationary branch and 5.5 times the first harmonic). The fluctuating part of the strain field carries the cost, and Eq. (26) makes that statement exact.

4. *The branch recovered by optimization is not the branch that minimizes dissipation at fixed control effort.* The most frequently recovered stationary structure is a net penalty at matched norm, and the most strongly time-dependent branch is the most dissipative of the three. Accessibility, reproducibility and physical performance are distinct properties of an actuation, and only the last is measured by the cycle-mean dissipation.

The practical implication for modal flow control is that a comparison of actuation strategies must state its constraint explicitly. Equal wall speed, equal quadratic norm and equal mechanical power define three different problems and can produce three different rankings; the present study uses the quadratic norm because it is the constraint available to an optimizer, and reports the peak velocity so that the effective Reynolds number of each comparison is visible. The cycle-mean dissipation, not the mean-field value and not the modal spectrum, is the performance index that follows from the energy identity.

Three extensions follow directly. A grid study of the time-dependent branches would establish whether the factor-of-three separation is resolution-independent to the same degree as the stationary ranking. Retraining under an explicit cycle-mean norm constraint, rather than rescaling an identified control after the fact, would separate the effect of the constraint from the effect of the identification path. Finally, three-dimensional cavities would introduce spanwise modes and test whether the two-cell second-harmonic state survives when the flow can vary along the span. The physics-informed identification remains, in this programme, the tool that generates the candidate actuations; the fluid-mechanical evaluation is what decides between them.

Data availability

The actuation series, the field captures and the post-processing scripts that support the figures and tables of this article are available from the corresponding author on reasonable request.

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References

[1] U. Ghia, K. N. Ghia, and C. T. Shin, "High-Re solutions for incompressible flow using the Navier–Stokes equations and a multigrid method," J. Comput. Phys. **48**, 387–411 (1982).

[2] O. Botella and R. Peyret, "Benchmark spectral results on the lid-driven cavity flow," Comput. Fluids **27**, 421–433 (1998).

[3] P. N. Shankar and M. D. Deshpande, "Fluid mechanics in the driven cavity," Annu. Rev. Fluid Mech. **32**, 93–136 (2000).

[4] S. Albensoeder and H. C. Kuhlmann, "Accurate three-dimensional lid-driven cavity flow," J. Comput. Phys. **206**, 536–558 (2005).

[5] H. C. Kuhlmann and F. Romanò, "The lid-driven cavity," in *Computational Modelling of Bifurcations and Instabilities in Fluid Dynamics*, edited by A. Gelfgat (Springer, Cham, 2019), pp. 233–309.

[6] M. Gad-el-Hak, *Flow Control: Passive, Active, and Reactive Flow Management* (Cambridge University Press, Cambridge, 2000).

[7] S. L. Brunton and B. R. Noack, "Closed-loop turbulence control: Progress and challenges," Appl. Mech. Rev. **67**, 050801 (2015).

[8] C. W. Rowley and S. T. M. Dawson, "Model reduction for flow analysis and control," Annu. Rev. Fluid Mech. **49**, 387–417 (2017).

[9] T. R. Bewley, P. Moin, and R. Temam, "DNS-based predictive control of turbulence: an optimal benchmark for feedback algorithms," J. Fluid Mech. **447**, 179–225 (2001).

[10] M. Bergmann, L. Cordier, and J.-P. Brancher, "Optimal rotary control of the cylinder wake using proper orthogonal decomposition reduced-order model," Phys. Fluids **17**, 097101 (2005).

[11] H. Choi, P. Moin, and J. Kim, "Active turbulence control for drag reduction in wall-bounded flows," J. Fluid Mech. **262**, 75–110 (1994).

[12] M. Quadrio and P. Ricco, "Critical assessment of turbulent drag reduction through spanwise wall oscillations," J. Fluid Mech. **521**, 251–262 (2004).

[13] T. C. Corke, C. L. Enloe, and S. P. Wilkinson, "Dielectric barrier discharge plasma actuators for flow control," Annu. Rev. Fluid Mech. **42**, 505–529 (2010).

[14] J. Rabault, M. Kuchta, A. Jensen, U. Réglade, and N. Cerardi, "Artificial neural networks trained through deep reinforcement learning discover control strategies for active flow control," J. Fluid Mech. **865**, 281–302 (2019).

[15] P. Garnier, J. Viquerat, J. Rabault, A. Larcher, A. Kuhn, and E. Hachem, "A review on deep reinforcement learning for fluid mechanics," Comput. Fluids **225**, 104973 (2021).

[16] M. Raissi, P. Perdikaris, and G. E. Karniadakis, "Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations," J. Comput. Phys. **378**, 686–707 (2019).

[17] S. Cai, Z. Mao, Z. Wang, M. Yin, and G. E. Karniadakis, "Physics-informed neural networks (PINNs) for fluid mechanics: A review," Acta Mech. Sin. **37**, 1727–1738 (2021).

[18] B. R. Noack, K. Afanasiev, M. Morzyński, G. Tadmor, and F. Thiele, "A hierarchy of low-dimensional models for the transient and post-transient cylinder wake," J. Fluid Mech. **497**, 335–363 (2003).

[19] P. J. Schmid, "Dynamic mode decomposition of numerical and experimental data," J. Fluid Mech. **656**, 5–28 (2010).

[20] K. Taira, C. W. Rowley, O. T. Schmidt, D. S. Luchtenburg, S. L. Brunton, and B. R. Noack, "Modal analysis of fluid flows: An overview," AIAA J. **55**, 4013–4041 (2017).

[21] D. Sipp and A. Lebedev, "Global stability of base and mean flows: a general approach and its applications to cylinder and open cavity flows," J. Fluid Mech. **593**, 333–358 (2007).

[22] S. Beneddine, D. Sipp, A. Arnault, J. Dandois, and F. Lesshafft, "Conditions for validity of mean flow stability analysis," J. Fluid Mech. **798**, 485–504 (2016).

[23] S. B. Pope, *Turbulent Flows* (Cambridge University Press, Cambridge, 2000).

[24] L. D. Landau and E. M. Lifshitz, *Fluid Mechanics*, 2nd ed. (Pergamon, Oxford, 1987).

[25] C. R. Doering and C. Foias, "Energy dissipation in body-forced turbulence," J. Fluid Mech. **467**, 289–306 (2002).

[26] D. d'Humières, "Multiple-relaxation-time lattice Boltzmann models in three dimensions," Philos. Trans. R. Soc. Lond. A **360**, 437–451 (2002).

[27] S. Chen and G. D. Doolen, "Lattice Boltzmann method for fluid flows," Annu. Rev. Fluid Mech. **30**, 329–364 (1998).

[28] Q. Zou and X. He, "On pressure and velocity boundary conditions for the lattice Boltzmann BGK model," Phys. Fluids **9**, 1591–1598 (1997).

[29] X. F. Pan, A. Xu, G. Zhang, and S. Jiang, "Lattice Boltzmann simulation of particle-laden turbulent channel flow," Comput. Fluids **35**, 1432–1444 (2006).

[30] P. J. Roache, "Perspective: A method for uniform reporting of grid refinement studies," J. Fluids Eng. **116**, 405–413 (1994).

[31] I. B. Celik, U. Ghia, P. J. Roache, and C. J. Freitas, "Procedure for estimation and reporting of uncertainty due to discretization in CFD applications," J. Fluids Eng. **130**, 078001 (2008).

[32] A. Beniaiche, "Physics-informed identification and independent numerical verification of conditional modal collapse in active control of lid-driven cavity flow," Phys. Fluids **38**, 103612 (2026). doi:10.1063/5.0351347